# Technical Report

## Power Electronics System

Candidate Number: **12245**

Group Business Design Project (Group 1: Inaglobe)

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# Summary

This report details the technical feasibility of developing a groundwater detection system to inform the location of borehole drilling sites, with the aim of increasing the chance of drilling a productive borehole by accurately locating groundwater aquifers. This includes a comparison of conceptual methods for locating groundwater aquifers and evaluates these against the technical requirements and customer needs.

The first part of this report details the background research and high-level concept evaluations for groundwater detection systems. In line with the original brief, the initial aim was to design a system which could scan the sub-surface geological layers in order to identify locations where a drilled borehole would produce a productive amount of water. After the decision to pivot the team focus away from drilling technology, development of the geophysical measurement system was split into new sub-systems. Therefore, the second part of this report focuses only on the design of the power supply sub-system for geophysical resistivity surveying equipment.

This overarching aim of this project is increasing access to groundwater for rural communities in developing countries. To this end, there is a focus on low-cost technology-appropriate solutions throughout.

Ultimately, from the first part of the study, the decision was taken to move forward with development of a DC resistivity system. This method was evaluated to be the most versatile and cost-effective solution to meet the customer's technical requirements and commercial needs. The outcome of the second part of this study was to develop a DC-DC flyback converter to generate the high voltages required for this method of geophysical surveying. Additionally, available battery technologies were evaluated against commercial and technical requirements; a deep-cycling flooded lead-acid battery was evaluated to best meet the needs of the customer.

# Contents

## Abbreviations

| | |
|---|---|
| SAVWA | South African Volunteer Workcamp Association |
| NGO | Non-Governmental Organisation |
| NGA | National Groundwater Archive (South Africa) |
| DWS | Department for Water and Sanitation |
| SADC | Southern African Development Community |
| FEM | Frequency Domain Electromagnetics |
| TEM | Transient Electromagnetics |
| DC | Direct Current |
| AC | Alternating Current |
| SNMR | Surface Nuclear Magnetic Resonance |
| MRS | Magnetic Resonance Sounding |
| GPR | Ground Penetrating Radar |
| EMI | Electromagnetic Interference |
| COTS | Commercial Off The Shelf |
| FLA | Flooded Lead Acid |
| DoD | Depth of Discharge |
| AGM | Absorbent Glass Mat |
| LFP | Lithium Ferro Phosphate |
| SMPS | Switch Mode Power Supply |
| MOSFET | Metal Oxide Semiconductor Field Effect Transistor |

# 1. Introduction

Geophysical surveys are a well-established and useful tool for siting boreholes in groundwater aquifers. They can inform the most likely place to drill a borehole for a sustainable water source. As discussed in the Commercial Viability Report (Section 3.1.3), commercially available instruments are engineered to work in the toughest physical environments with high precision and accuracy. They are often **over-engineered** for many groundwater exploration use-cases, particularly in developing countries where simpler instrumentation would suffice.

The most significant barriers preventing wider uptake of hydrogeophysical work in developing countries, in order, are:

1. **Access to equipment**,
2. Access to software, computer hardware, and the internet,
3. Practical training.

This report will focus on improving access to geophysical equipment by developing **open-source**, **low-cost**, **appropriately engineered** equipment.

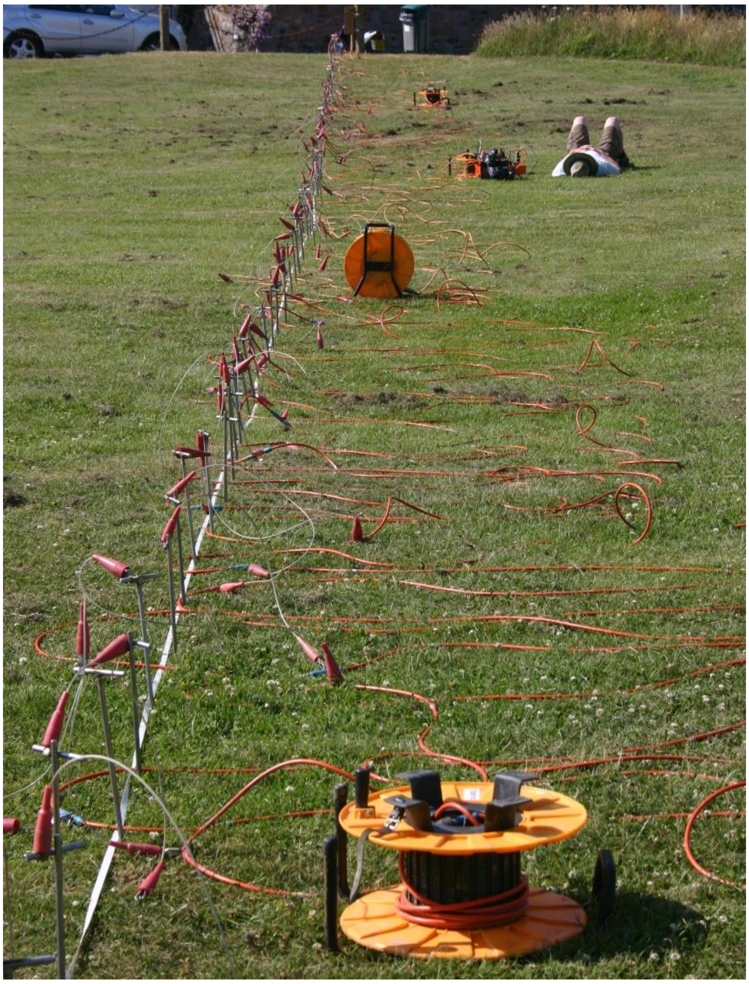


Figure 1 Example of a geophysical ERT (Electrical Resistivity Tomography) set up [31]

# 2. Problem Definition

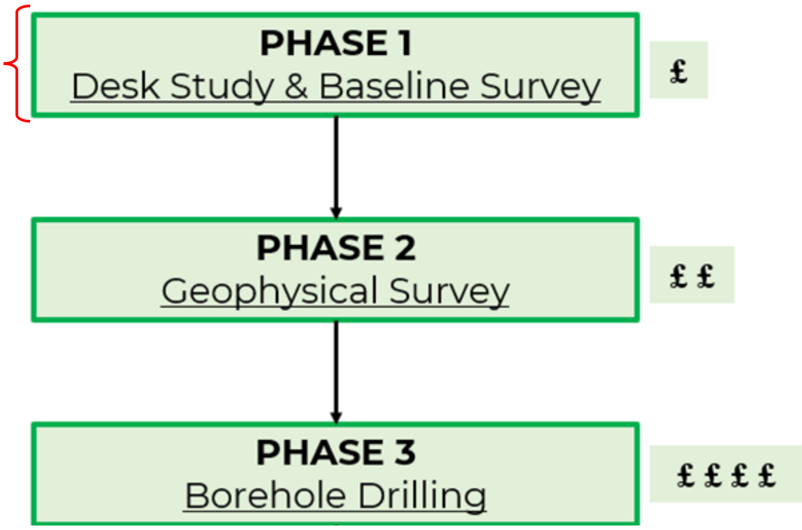



Figure 2 Stages of the groundwater exploration process [5]

## 2.1. Geological Context

When conducting groundwater exploration, it is crucial to have a good conceptual model of the local hydrogeology. This includes:

- Average depth to groundwater
- How does groundwater exist in the sub-surface?
- What kind of aquifer is it?
- What are the lithological layers?

With respect to designing geophysical surveying equipment, this information informs the choice of surveying method, allows better interpretation of the survey data, and can help gauge the expected sustainability of the well.

Figure 2 shows a simple breakdown of the groundwater exploration process. Phase 1, the desk study, includes studying the hydrogeological context. This phase incurs the least cost and can be used as a stage-gate to decide whether the project goes ahead or not. Careful consideration and a good understanding of context here can save money as the project progresses. More detail on this process can be found in the project management report [3].

As an example, hydrogeological data from the Soshanguve area is shown below. This background context will be used to inform the design requirements and concept/method selection later.

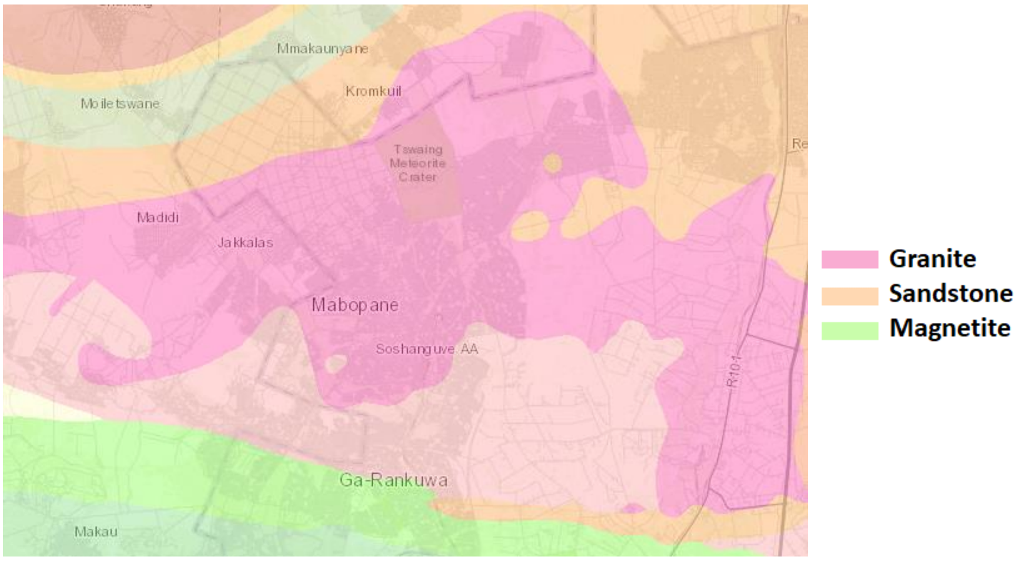
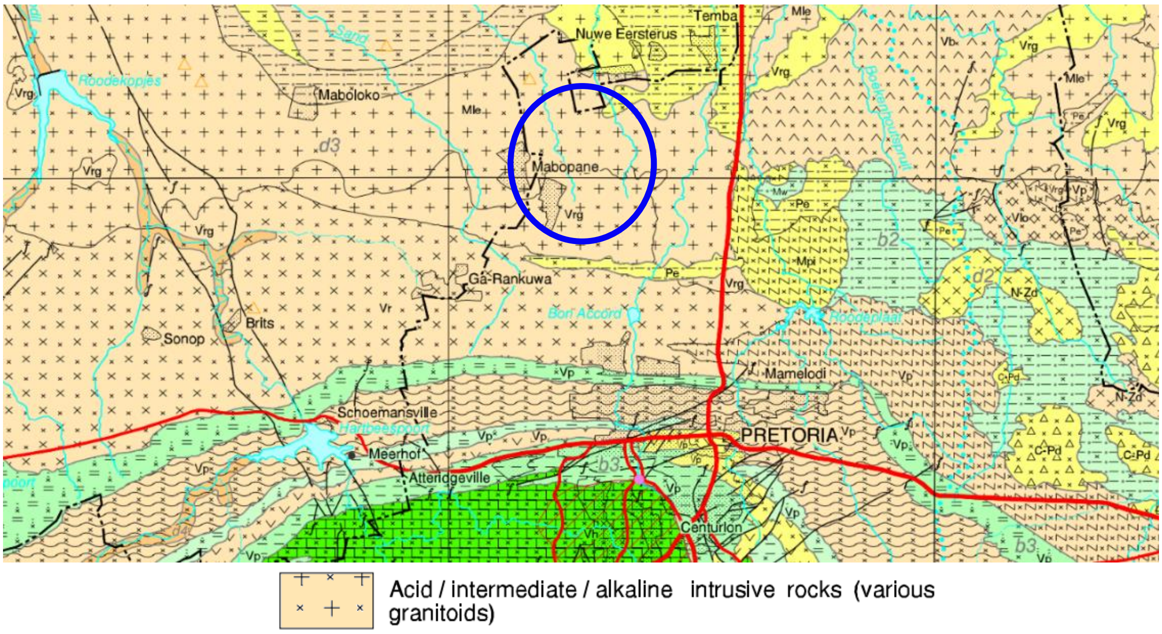



Figure 3 Geological map showing main lithology types in the area. [6]




Borehole yield (L/s)

| Aquifer Type | 0.0 - 0.1 | 0.1 - 0.5 | 0.5 - 2.0 | 2.0 - 5.0 | > 5.0 |
|---|---|---|---|---|---|
| Intergranular | * | * | * | * | a5 |
| Fractured | * | b2 | b3 | b4 | b5 |
| Karst | * | c2 | c3 | c4 | c5 |
| Intergranular and fractured | * | d2 | d3 | d4 | d5 |

Figure 4 Hydrogeological context of the area. SAVWA operates in the area highlighted by the blue circle. This area is characterised by low productivity (<2 L/s) fractured granitoid aquifers. [7]

Figure 3 shows the lithology (rock type) in the area. Soshanguve is dominated by igneous granitic rocks such as magnetite and granite, with sandstone deposits to the north. Similarly, Figure 4 is a hydrogeological map of the area which simultaneously shows lithology (hatching), but also aquifer type and productivity (colour). The Soshanguve area is characterised by **intergranular and fractured aquifers** occurring in **granitoid intrusive rock** formations. Expected borehole **yields are quite low** (0.5-2 Litres/s) but can be enough to support a household or small community well.

Figure 5 and Figure 6 depict lithological layers present in the context, including a top layer of highly weathered rock or soil underlain by fractured basement rock. The basement rock become less weathered and fractured with depth.

Below the water table, it is only in these fractures that groundwater is found. The granitic basement rock found in the area is highly impermeable, meaning water cannot flow through the rock or exist intergranularly. This means that boreholes can be drilled hundreds of metres deep and will not produce viable amounts of water if they do not intersect enough fractures.

Figure 5 and Figure 6 also show deep geological faults which are areas of deeper fracturing. These are the target for borehole drilling in this geology, as they yield more water and are constitute a sustainable freshwater source.

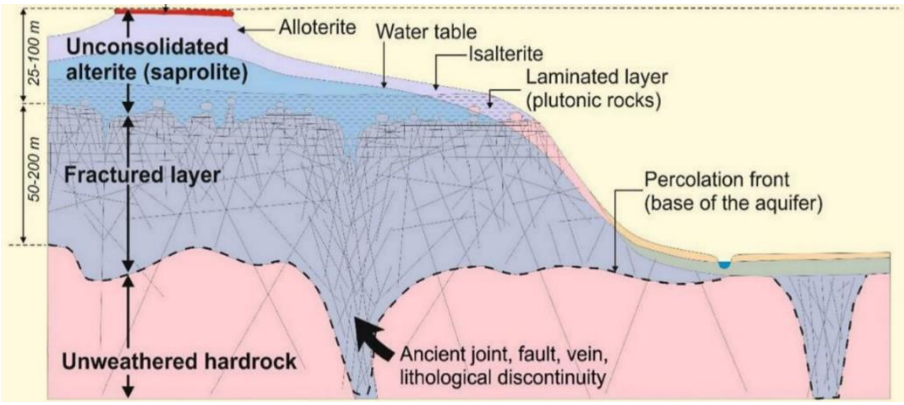



Figure 5 Conceptual model of hydrogeological structures in weathered basement rock [8]

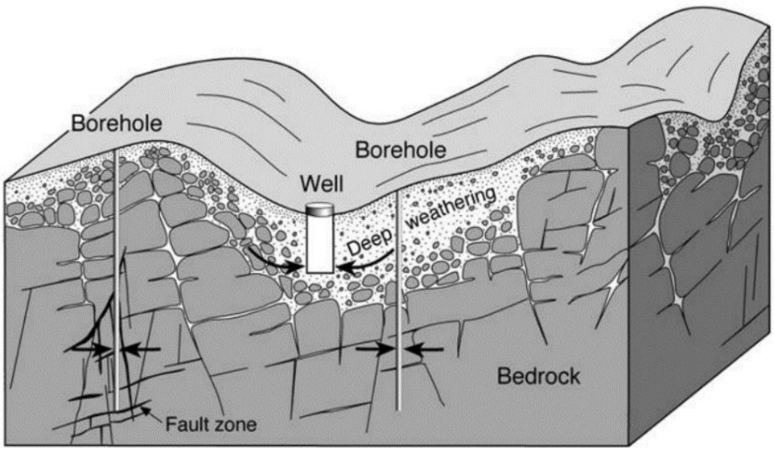



Figure 6 Groundwater occurrence in weathered basement rock [9]

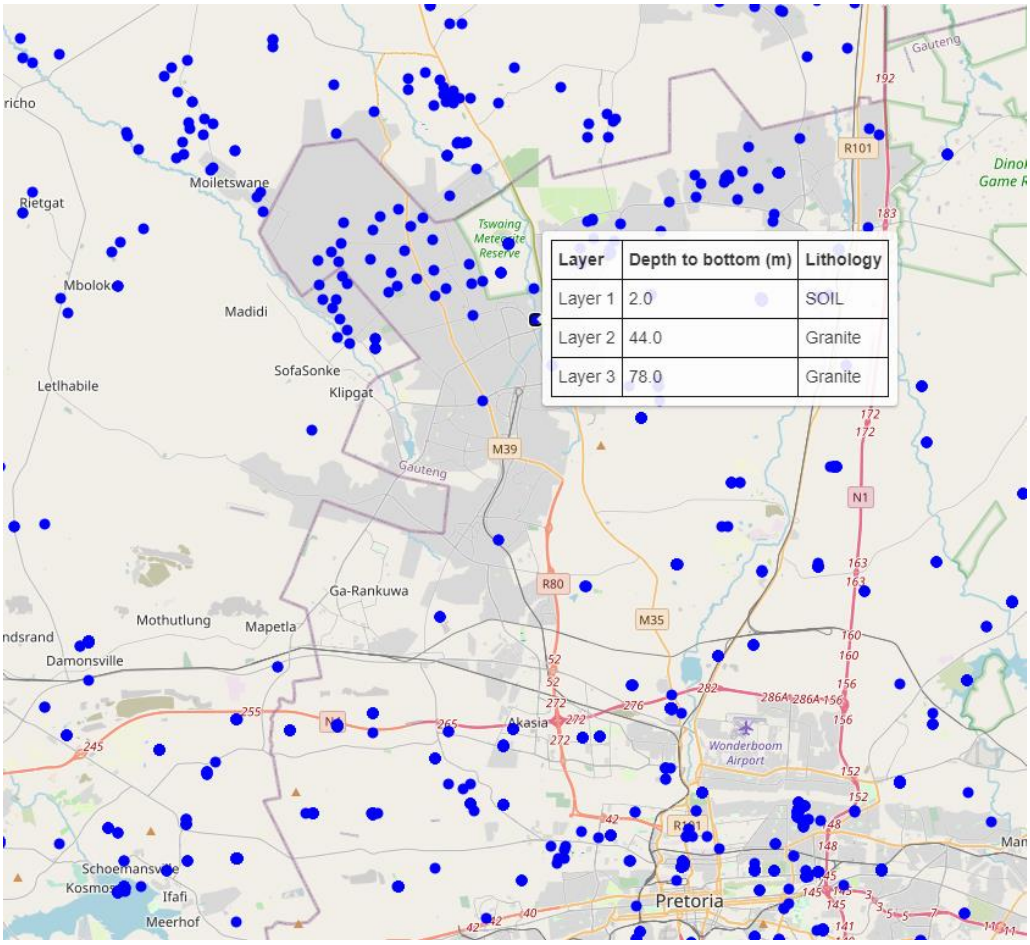



Figure 7 Custom interactive map of registered boreholes in the area showing lithology layers and associated depths

Figure 7 shows a custom interactive map designed for this project. This uses freely available data from 300,000+ boreholes in the South African National Groundwater Archive (NGA) [10]. At each borehole location lithology layers and corresponding rock type are plotted.

A caveat to this is that most boreholes are **not** registered – especially domestic private boreholes. However, this tool is useful for confirming local rock types and also indicating borehole depths in the area; giving an initial estimate of the depth required to produce a viable amount of water. This data is available for

This map can be viewed at: [Interactive_map_Soshanguve.html]

Associated code is in Appendix A, and data sets are attached as supporting documents.

## 2.2. Customer Needs

Before exploring potential solutions, a preliminary specification based on high-level fundamental requirements of the system were set to guide the design process. The customer needs of the system have been explored in greater detail in the Commercial Viability Report [2]. Table 1 shows some of the customer needs of the geophysical survey system.

Table 1 Preliminary Requirements Specification for groundwater exploration system

| Requirement: | Must/Wish |
|---|---|
| System can identify the most likely location for a productive borehole within the scanned area | Must |
| System can scan a target area within a reasonable amount of time (e.g.: 4-5 km in 2-3 days) | Wish |
| Data collection process does not require a hydrogeologist on-hand | Must |
| Post-processing and analysis of data does not require a hydrogeologist | Wish |
| System should be low-cost but value for money | Must |
| Technology should be locally repairable (easy maintenance) | Must |

# 3. Concept Comparison

Research into geophysical methods for groundwater exploration showed that they can be broadly grouped as: **electrical**, **electromagnetic,** or **seismic** methods.

Table 2 summarises some of the main techniques for groundwater exploration. Sections 3.1 to 3.3 compare the various conceptual geophysical methods and the theory behind them.

Table 2 Comparison of the most common methods for locating groundwater reservoirs [9]

| Geophysical Technique | What is Measured? | Output | Depth of Investigation (m) | Positives | Negatives |
|---|---|---|---|---|---|
| **TEM** (Transient Electro-Magnetics) | Apparent electrical resistance | 1D resistivity sounding | 300 | Better depth of investigation than FEM. | Expensive and difficult to operate |
| **SNMR** (Surface Nuclear Magnetic Resonance) | Water content of sub-surface | 1D sounding of water content with depth as a % | 150 | Directly measures water content of the sub-surface | Extremely expensive, highly technical, excessive bulky equipment |
| **DC Resistivity** | Apparent electrical resistivity | 1D: (VES) vertical geoelectric section<br><br>2D/3D: (ERT) Pseudo-section of sub-surface resistivity | 100 | Determines thickness of the weathered zone. Identify differences in geology. | Relatively slow survey and requires careful interpretation.<br>Very long wires needed for deep survey (300m+) |
| **FEM** (Frequency Domain Electro-Magnetics) | Apparent electrical conductivity | 1D: Single transect ground conductivity.<br><br>2D: contoured surface of bulk ground conductivity | 50 | Very quick and easy survey. Determines thickness of weathered zone. good for identifying fracture zones | Careful interpretation required.<br>Compact survey equipment |
| **Seismic Refraction** | P-wave velocity | 2D section of P-wave velocity | 30 | Can locate fracture zones in basement and thickness of sediments. | Quite slow and difficult to interpret data |

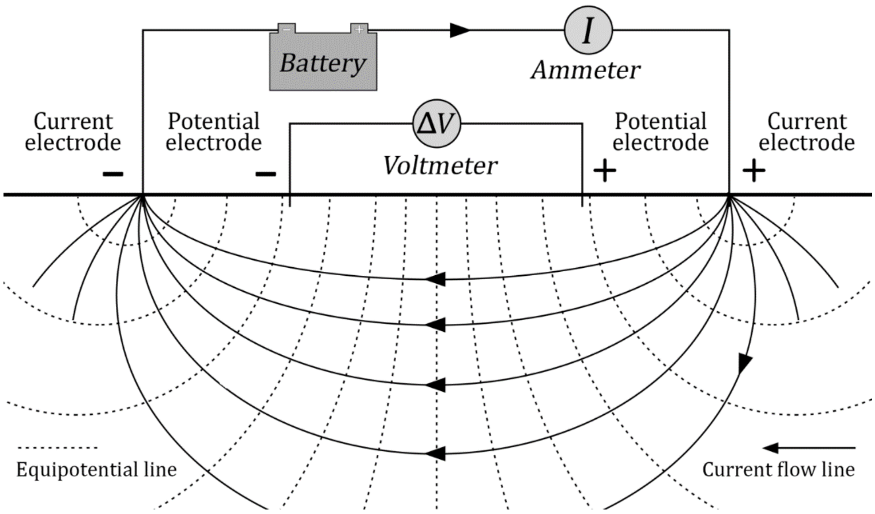



Figure 8 Electrical resistivity diagram [11]

## 3.1. Electrical Resistivity

Resistivity surveys involve passing a continuous electrical current into the ground through a pair of current electrodes and then measuring the voltage distribution across the surface of the ground using a pair of potential electrodes (Figure 8). Using Ohm's law, the resistance of the ground can be calculated, which can be used to determine the fundamental property **resistivity** of the ground by making a few assumptions.

Figure 9 shows how resistivity is calculated from resistance. Resistance is dependent on length and cross-sectional area, while resistivity is unchanged. Calculating resistivity from the surface is not straightforward, however. It requires a numerical modelling technique called inversion to estimate resistivities at certain depths based on the electrode configuration. This is explored more in Jack's report [4].

$$\boldsymbol{R = \frac{\rho L}{A}}$$

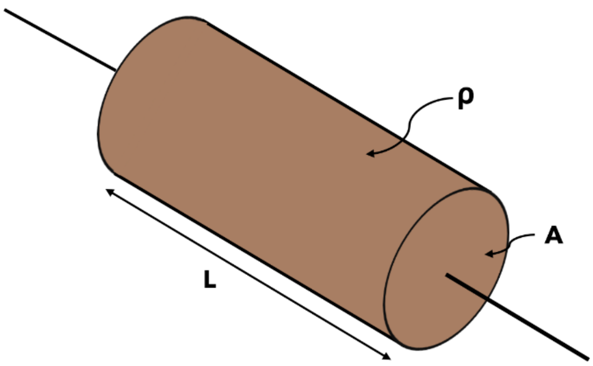



Figure 9 Relationship between resistance and resistivity

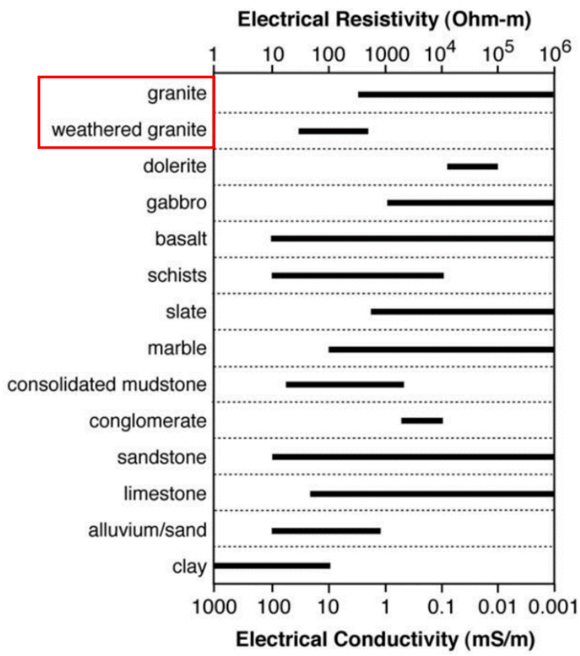


Figure 10 Electrical resistivity values for common lithologies containing groundwater [9]. Granite is highlighted as an example as this is the local basement lithology in Soshanguve

Once the apparent resistivities have been calculated they can be compared to values in Figure 10, which shows the ranges of resistivities for many common rock types. Weathered granite with water in fractures is much less resistive than unweathered granite; this method will indicate good places to drill by finding the lowest resistivity area.

Resistivity surveys can be conducted with different electrode configurations (arrays), allowing for 1-D soundings, 2-D profiles, or 3-D interpretations of the subsurface. Increasing technical complexity and computational power is incurred with more detailed data.

1-D Resistivity surveys can be done in two different ways:

- **Horizontal Electrical Profiling (HEP)** (Figure 11) measures resistivity along a transect. The electrode setup is moved along a transect to sample resistivities at many locations to **qualitatively** estimate where the weathered zone is deepest.

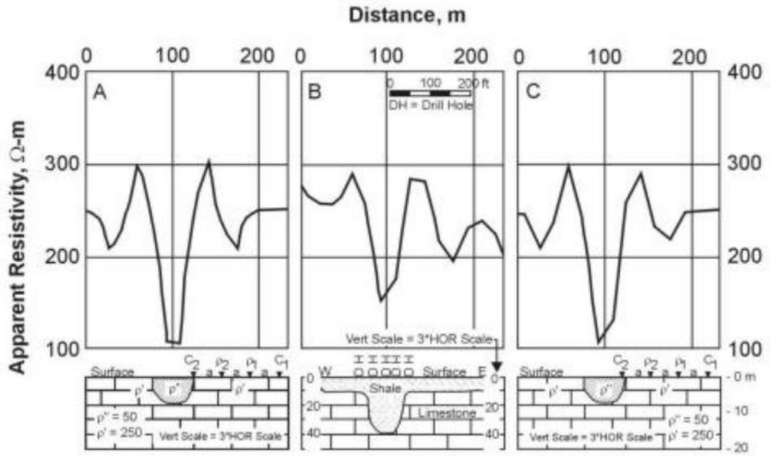


Figure 11 Example HEP data over 3 theoretical geological structures [12]

- **Vertical Electrical Sounding (VES)** measures changes in resistivity with depth at a single location by taking many measurements while increasing the electrode spacing. Figure 12 shows example sounding curves which are taken by gradually increasing electrode separation to sample deeper section of the sub-surface [13].

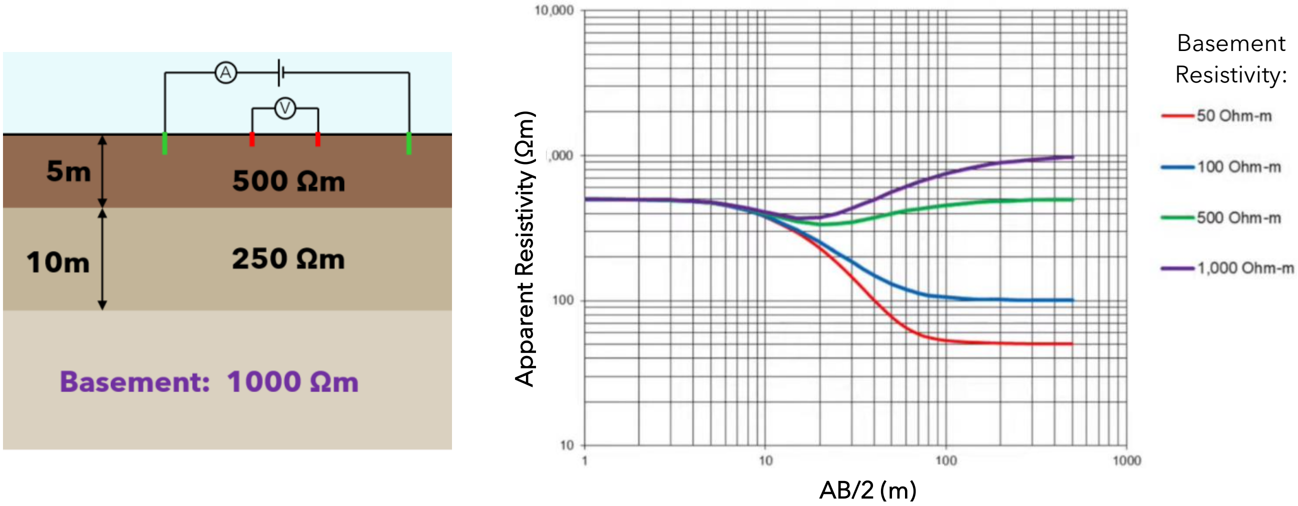



Figure 12 Example VES data. Left: lithological layers with varying resistivities. Right: Sounding curves taken by increasing electrode spacing. Each curve represents a different basement resistivity [13]

2-D and 3-D surveys are done by setting up a grid of electrodes and taking many combinations of electrodes as current injectors or potential measurers to build a very detailed map of sub-surface resistivity. Comparing these methods against 1-D resistivity has been done more extensively in Jack's report [4]. 2/3-D surveys are overkill in the context of the project and 1-D surveys are wholly sufficient.

## 3.2. Electromagnetics

Various techniques using electromagnetics (EM) are used for groundwater exploration. These methods all involve emitting a form of EM pulse and measuring the reflected feedback from sub-surface.

### 3.2.1. Frequency EM

FEM measures the electrical conductivity of the ground by transmitting a sinusoidally varying current at a specific frequency [9]. Figure 15 demonstrates the principle of this method. Both the primary and induced secondary EM fields are measured by the receiver coil. Apparent conductivity can be found by assuming a linear ratio of secondary:primary field strength. The measured apparent conductivity is a weighted average of the ground within the range of investigation.

The benefit of FEM is that it does not require electrical connection to the ground, so it is much quicker to set up than DC resistivity surveys [15]. In many cases FEM surveys have replaced DC resistivity profiling due to the speed and ease of scanning a large area quickly. FEM is more compact and lightweight, making it easier to transport, requires just 2 operators, and needs <40m of space for each measurement.

The disadvantage is that minimal vertical-sounding data is collected compared to VES. Since FEM measures apparent conductivity averaged with depth, interpretation is usually **qualitative** and cannot determine variations with depth – similar to HEP [18].

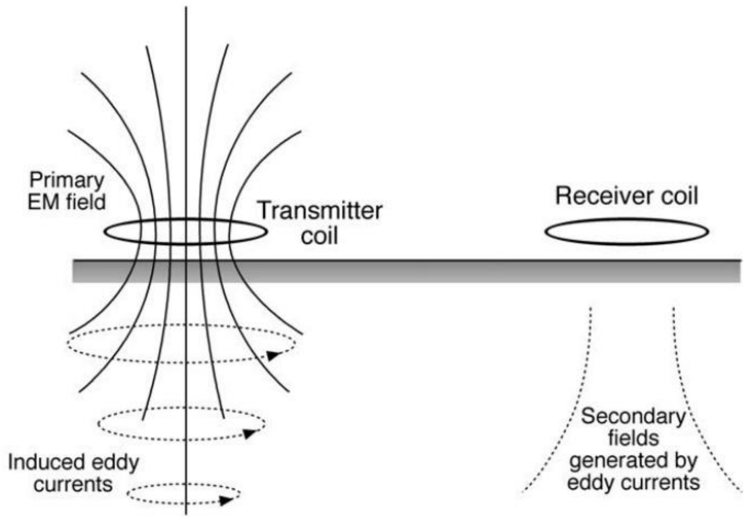



Figure 13 Principle of frequency domain EM (FEM) geophysical survey. Two separate coils, Transmitter and Receiver, are required to induced and measure eddy currents, respectively [9]

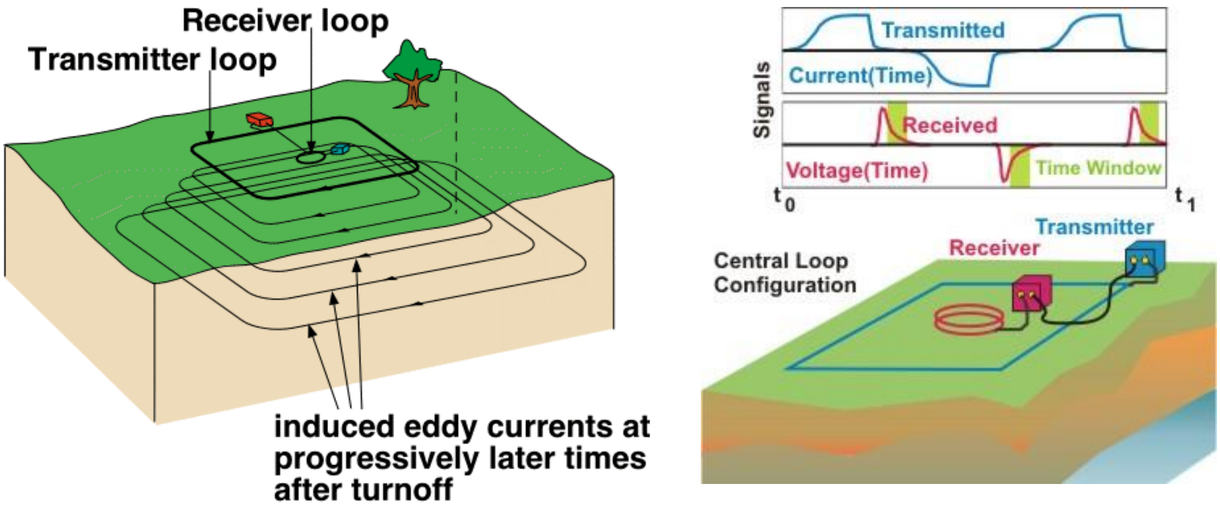



Figure 14 Principle of TEM surveys. Left: distribution of eddy currents on transmitter current turn-off. Right: Transmitter vs Receiver signal showing how the collapsing magnetic field is measured by the receiver [17]

### 3.2.2. Transient EM

TEM uses the transient decay of an induced secondary field in the ground to calculate resistivity. Figure 16 demonstrates the principle of this method. The current in the transmitter loop is switched off abruptly collapsing the magnetic field, inducing eddy currents. The receiver loop measures the induced secondary fields that are generated in the few milliseconds after the transmitted signal transition. As with FEM surveys, each TEM measurement is a vertical 1-D sounding. TEM systems can up to 400 metres deep [9][16].

The benefit of TEM for sounding measurements over DC resistivity is that it takes up less space and can measure deeper. A 40m square loop can measure 200-250m depth, while a DC resistivity survey would require over 600m of electrode separation for the same depth.

However, TEM systems are expensive and difficult to operate. TEM is not routinely used for groundwater surveys in Africa/India for this reason, despite its technical superiority.

### 3.2.3. Surface Nuclear Magnetic Resonance (SNMR)

SNMR is the only method that can directly measure water content of the sub-surface at different depths without physically drilling into the ground [20]. SNMR involves a large loop of cable on the surface which acts as both transmitter and receiver. Alternating current at the harmonic frequency of protons is passed through the transmitter which generates an EM field in the sub-surface. This specific frequency offsets the axis of the quantum spin of protons from equilibrium under. Upon switching off the transmitter, the proton spins realign with Earth's magnetic field in unison, inducing a voltage in the receiver. The amplitude of the signal is proportional to the density of protons, which directly correlates to the water content in the scanned area [19].

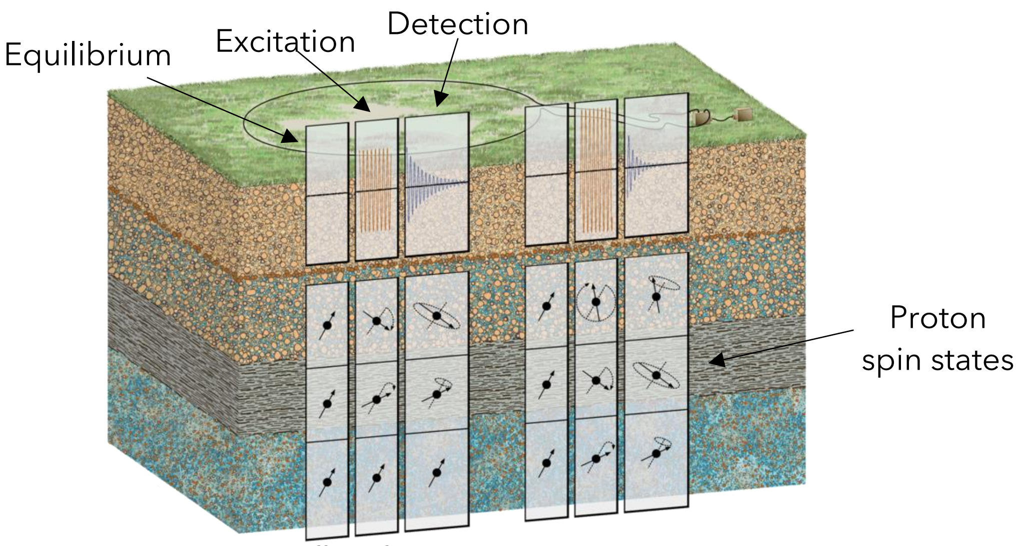



Figure 16 Diagram highlighting the effect of SNMR pulse on proton spin. Spin is originally aligned with Earth's magnetic field, then offset by the transmitted pulse, before oscillating back to equilibrium [21][22]

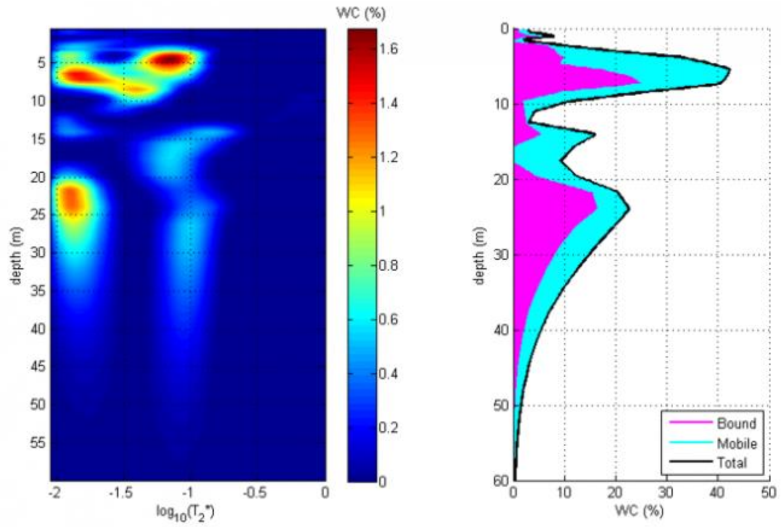



Figure 15 SNMR survey data. Left: decay time indicating porosity with depth. Right: Water content (%) with depth

Figure 17 shows the 3 stages of the SNMR measurement. Initially protons are aligned with Earth's magnetic field before they are offset by the artificially induced field. Upon switching off the transmitter the protons oscillate back to equilibrium – generating a measurable signal.

Figure 18 shows that SNMR is uniquely superior to other geophysical methods in that it **directly** measures water content of the sub-surface, while other methods can only infer based on physical characteristics (resistivity/conductivity).

The disadvantage of SNMR in the context of this project is that it is highly technically advanced and requires lots of heavy specialist equipment – making it hard to transport.

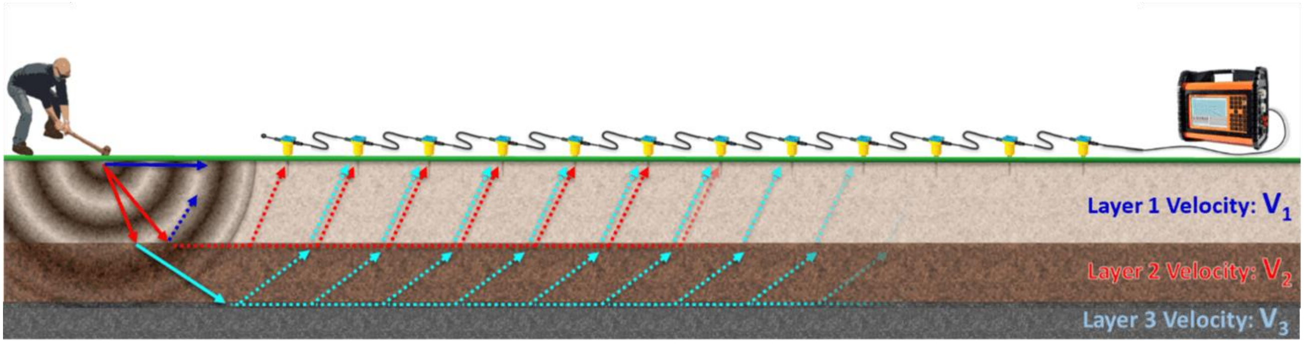



Figure 17 Principle of seismic refraction survey. P-waves have different velocities through different mediums [23]

## 3.3. Seismic Refraction

Seismic refraction uses an array of vibration sensors to pick up refracted P-wave vibrations from an excitation source. P-waves travelling through the sub-surface are refracted from layer interfaces. Geophones detect the arrival time of the refracted waves to determine the geological layer thickness and composition. This can be interpreted to identify the depth to bedrock. [24]

This method is limited in depth (<30m), but it can be useful for determining the thickness of the weathered zone.

# 4. Discussion

## 4.1. Concept Evaluation

Criteria used to evaluate the groundwater exploration methods have been weighted in a pairwise comparison (Table 4). Criteria were chosen based on customer requirements of the system[1][2]. A multi-choice decision analysis (Table 5) was then performed to rank each concept in the context of the project.

From this analysis, the most suitable geophysical exploration method is a 4-electrode DC resistivity survey. The quality and usefulness of data from TEM and SNMR is unmatched, but these methods are still highly technical, extremely expensive, and completely unsuitable in the context of this project.

FEM is a very quick and lightweight but is largely a qualitative method and determining variations with depth is difficult. Often FEM is used in conjunction with DC resistivity. This is disadvantageous in this context as it would require development of two separate geophysical instruments.

Seismic refraction is the only method that can match DC resistivity's low-cost. However, seismic methods are limited in depth and the quality of data produced.

Overall, the most suitable method for this project is DC resistivity. This method best fits with the project brief in making groundwater accessible for rural communities in developing countries by utilising commonly available low-cost technology. Furthermore, DC resistivity can be used for both profiling (HEP) and sounding (VES) measurements, so no other geophysical instruments are required.

## 4.2. Development Plan and Change of Scope

As detailed in Arthur's report [3], the project experienced a change of scope halfway through. As a result, all team members were assigned onto development of the geophysical survey equipment as per Table 3.

**A shortened feasibility study follows this focussed on the design decisions within the power supply sub-system.**

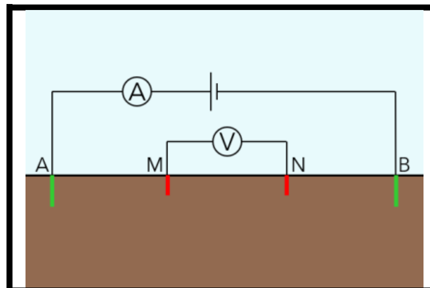
Table 3 Reassignment of team roles following change in project focus.

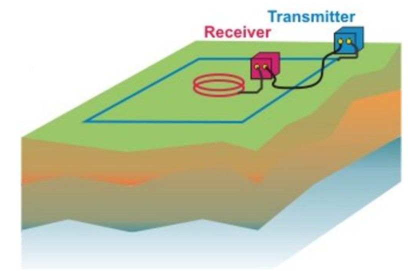
| System Breakdown | Sub-System | System Lead |
|---|---|---|
| Project Management | Team Management<br>Business Lead<br>System Integration | Arthur Anstis |
| Power Electronics & Power Source | Power Electronics<br>Power Source<br>PCB Design | Harry Mills |
| Data Processing & App Design | Digital UI<br>Data Processing<br>Data Analysis & Output | Jack Parsons |
| Electrode Design & Process Optimisation | Electrode work package<br>Process Optimisation<br>Transportability | Daniel Lipscombe |
| Signal Processing and Circuit Design | Bluetooth<br>Signal Processing<br>PCB Design | Alex Smith |
| Hardware Casing and Physical UI | Hardware Casing<br>Manufacturability<br>Physical UI & Feedback | Caro Ashforth |

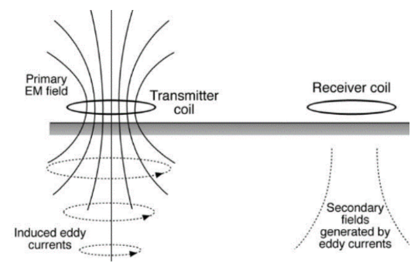
Table 4 Pairwise comparison of evaluation criteria for ranking concepts

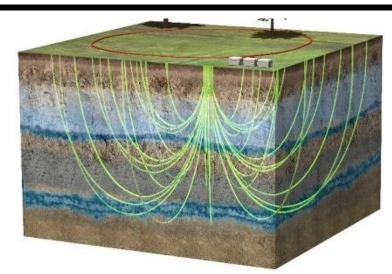
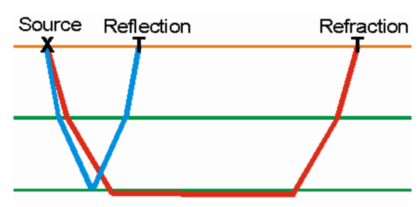
| | Data value | Technical simplicity | Usability/ training required | Low-cost | Speed of taking measurements | Size/ moveability | Depth of investigation | Score | Weight |
|---|---|---|---|---|---|---|---|---|---|
| Data value | x | 0 | 0 | 0 | 1 | 1 | 1 | 3 | 6 |
| Technical simplicity | 1 | x | 1 | 0 | 1 | 1 | 0 | 4 | 7.5 |
| Usability/training required | 1 | 0 | x | 0 | 0 | 1 | 1 | 3 | 6 |
| Low-cost | 1 | 1 | 1 | x | 1 | 1 | 1 | 6 | 10.5 |
| Speed of taking measurements | 0 | 0 | 1 | 0 | x | 0 | 0 | 1 | 3 |
| Size/moveability | 0 | 0 | 0 | 0 | 1 | x | 0 | 1 | 3 |
| Depth of investigation | 0 | 1 | 0 | 0 | 1 | 1 | x | 3 | 6 |

Table 5 MCDA matrix scoring each method against the weighted criteria. DC resistivity is the most suitable method to move forward with








| | | DC Resistivity | | TEM | | FEM | | SNMR | | Seismic | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Criteria | Weighting | Score/5 | Weighted Score | Score/5 | Weighted Score | Score/5 | Weighted Score | Score/5 | Weighted Score | Score/5 | Weighted Score |
| Low-cost | 10.5 | 5 | 52.5 | 2 | 21 | 3 | 31.5 | 1 | 10.5 | 5 | 52.5 |
| Technical simplicity | 7.5 | 5 | 37.5 | 2 | 15 | 3 | 22.5 | 1 | 7.5 | 4 | 30 |
| Usability/training required | 6 | 3 | 18 | 3 | 18 | 5 | 30 | 1 | 6 | 4 | 24 |
| Data value | 6 | 3 | 18 | 4 | 24 | 3 | 18 | 5 | 30 | 1 | 6 |
| Depth of investigation | 6 | 4 | 24 | 5 | 30 | 2 | 12 | 4 | 24 | 1 | 6 |
| Speed of taking measurements | 3 | 1 | 3 | 2 | 6 | 5 | 15 | 2 | 6 | 4 | 12 |
| Size/moveability | 3 | 4 | 12 | 2 | 6 | 5 | 15 | 2 | 6 | 4 | 12 |
| | TOTAL SCORE: | | 165 | | 120 | | 144 | | 90 | | 142.5 |

# 5. Power Sub-System Problem Definition

Figure 19 shows a diagram of a typical system with the key components highlighted:

- Battery (usually 12V)
- DC-DC converter (12V → high voltage)

## 5.1. Battery Power

In developing countries the only available option is usually a 12V car or motorbike battery. For example, Figure 20 shows a 12V 40Ah lead-acid battery available in South Africa for R1250 (£50). In the interest of this study, newer battery architectures will also be considered despite practical problems sourcing the technology in developing countries.

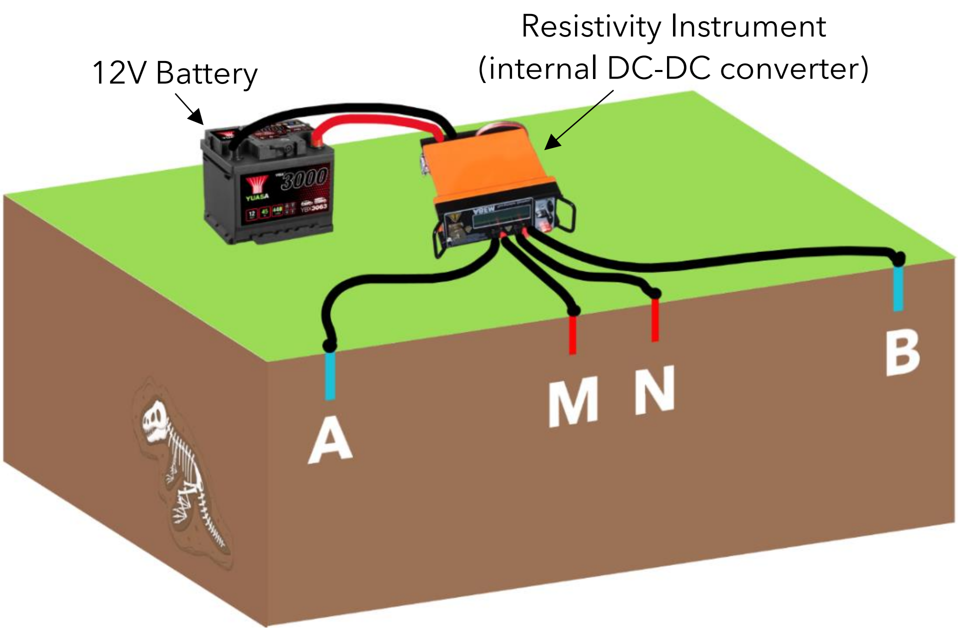



Figure 18 Experimental set up of a DC resistivity survey highlighting power supply sub-system components


Figure 19 Typical battery used in geophysical surveys: 12V 40Ah lead-acid car battery available in South Africa

## 5.2. Why is high voltage required?

### 5.2.1. Equivalent Circuit

DC-DC conversion from battery 12V DC up to several hundred volts are often required to generate measurable potential differences across the MN electrodes.

Figure 22 shows the equivalent circuit of a 4-electrode survey with typical resistance values in Table 8. The contact resistance between the electrodes and ground dominates the resistance of the system. The total resistance path through the ground is given by equation 1.

$$2\,R_{contact} + R_{AM} + R_{MN} + R_{NB} = R_{Total} \tag{1}$$

For the example data in Table 8, $R_{total}$ = 2015Ω. Assuming $V_{AB}$ = 500V the voltage drop across MN can be found.

$$\frac{R_{MN}}{R_{Total}} = \frac{1}{2015} = 0.05\% \tag{2}$$

$$V_{MN} = V_{AB} \cdot 0.05\% = 0.25V \tag{3}$$

The power supply specifications of some commercial instruments are shown in Table 9.

Table 6 Typical equivalent resistances within a DC resistivity survey [13]

| Resistance | Typical value |
|---|---|
| $R_{contact}$ | 1000 Ω |
| $R_{AM}$, $R_{MN}$, $R_{NB}$ | 1 Ω |
| $R_{input}$ | 100 MΩ |

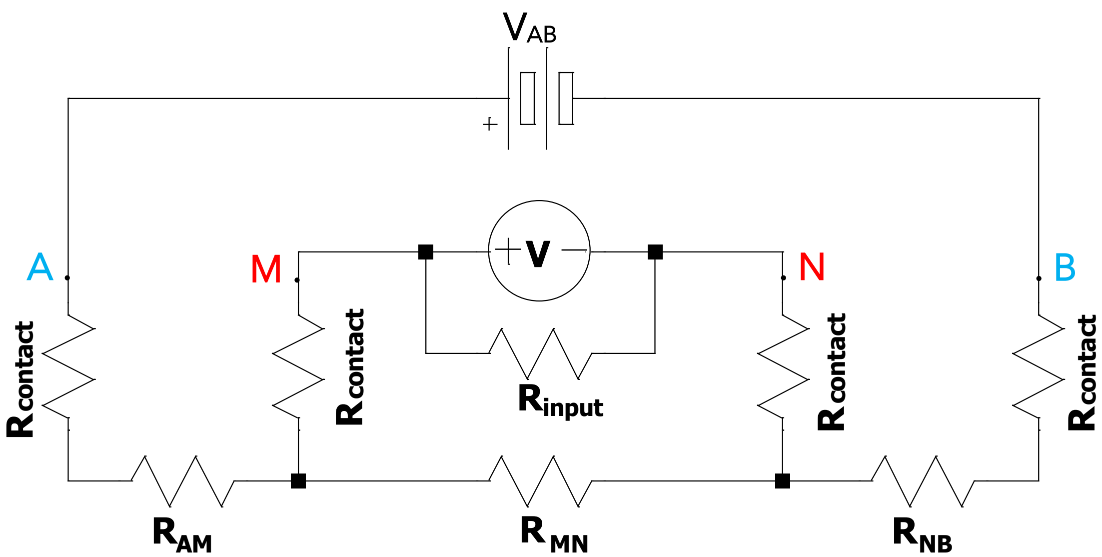



Figure 20 Equivalent circuit schematic of a 4-electrode DC resistivity survey

# 6. Power Sub-System Requirements Specification

Table 7 shows the requirement specification for the Power Supply sub-system. Requirements were set using guidance from Pahl and Beitz (2007)[33] and based on high-level customer needs in the CVR [2].

Table 7 Power Supply Sub-System Requirements Specification

| | Requirement | Target | Must/ Wish | Method of Evaluation |
|---|---|---|---|---|
| 1 | Battery | | | |
| 1.1 | Battery can power the system for multiple days of survey work | Minimum 20 Ah capacity | Must | Check specification |
| 1.2 | Battery can deliver enough current to power the DC-DC converter | >30A | Must | Check specification |
| 1.3 | Battery does not discharge to unsafe levels during use | Power supply shuts off when battery voltage is too low | Must | Circuit design or user interface |
| 1.4 | Battery can be recharged easily | Safely charges overnight | Must | Technology analysis |
| 1.5 | Battery internals are non-hazardous and can be disposed of safely | Prefer non-toxic batteries | Wish | Technology analysis |
| 1.6 | Battery can operate in suitable temperature range | -10°C to 40°C | Must | Check specification |
| 1.7 | Battery does not need regularly replacing | Expected life >1 year | Must | Check specification |
| 1.8 | Battery connections are secure and failsafe | Battery cannot be connected in reverse polarity. Terminals and sockets are clearly marked | Must | Design verification |
| 1.9 | Battery should be low-cost | Suitable technology in developing countries preferred | Wish | Cost comparisons |
| 1.10 | Battery can be carried comfortably by a single person | Less than 15kg | Wish | Check specification |
| 2 | DC-DC Converter | | | |
| 2.1 | Can reliably convert battery voltage to required output voltage | Minimum 500V output | Must | Circuit simulation/ prototyping |
| 2.2 | Converter can drive enough power to reach required voltage | 100W output power | Must | Circuit simulation/ prototyping |
| 2.3 | Converter includes external on/off switch | Digitally controlled power on/off | Wish | Design verification |
| 2.4 | Efficiency is not so low battery power is rapidly depleted | >70% conversion efficiency in normal use | Wish | Circuit simulation/ prototyping |
| 2.5 | Output polarity can be switched to avoid ground polarisation | H-bridge or digitally controlled switching | Wish | Design verification |
| 2.6 | Technology is low complexity | Manufactured/maintained in developing countries | Wish | Design verification |
| 2.7 | Safety mechanisms are included to mitigate operator risk | Short circuit protection | Must | Check relevant safety regulations |
| 2.8 | Electromagnetic Interference (EMI) emissions are minimised | EMI shielding on key parts of the system | Wish | EMI compliance regulation tests |
| 2.9 | Output voltage ripple is minimised | <1V ripple | Wish | Circuit simulation/ prototyping |

# 7. Power Sub-System Concept Comparisons

## 7.1. Battery Architectures

In DC resistivity surveys measurements are taken by injecting current into the ground for a few seconds. Low-end professional equipment often has a maximum power of 100W. Therefore a 12V battery would need to provide 8-10A in short bursts (2-4 seconds). An initial estimate of the required capacity can be calculated assuming 3 HEP surveys with 200 measurements each are run per day:

$$Q = 10 \cdot \frac{4}{3600} \cdot 200 = 6.6\ Ah/day \qquad (4)$$

Deep-cycling regular car batteries below 80% can rapidly degrade them, so a 40Ah battery would need to be checked and charged every couple of days.

### 7.1.1. Flooded Lead-Acid (FLA)

Figure 24 shows the construction of a FLA battery. In each cell, a lead-alloy cathode and anode are submerged in electrolyte and prevented from touching with a separator [27].

Standard FLA batteries are starters for combustion engines which require bursts of very high power for short periods of time and are designed for low discharge rates as the alternator will quickly recharge the battery.

More expensive FLA batteries may be designed for 'deep-cycling'. These have thicker lead plates which have less surface area (and less instantaneous power) than standard FLA, which allows them to discharge as low as 20% of total charge repeatedly without damaging the plates [25]. Regularly cycling a standard FLA battery to 50% charge can reduce its life to just a few months (whereas a deep-cycle battery would last years).

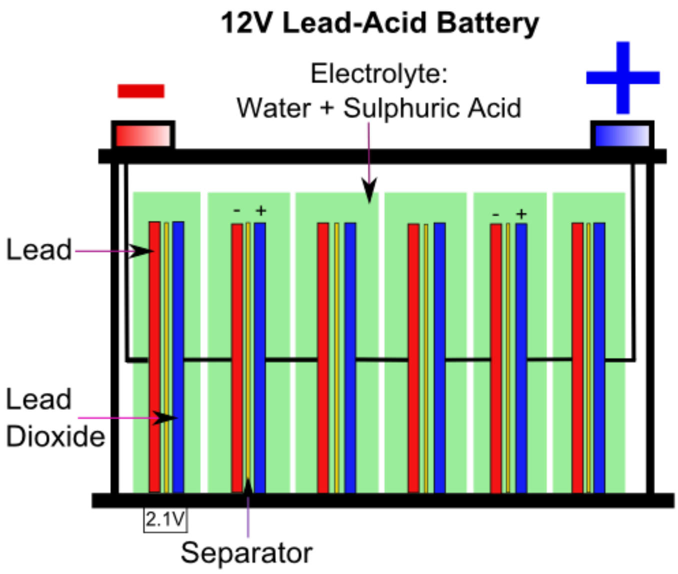



Figure 21 Diagram of a standard flooded lead-acid battery. Each cell provides 2V, so 6 are required for 12V output


Figure 22 Left: AGM battery, completely sealed and spillproof. Right: A bank of LFP batteries for a marine application with a custom battery protection module [30]

### 7.1.2. AGM

Absorbed Glass Mat batteries are another type of deep-cycling lead-acid batteries. These operate on the same principle as flooded lead-acid, except they use Boron-silicate glass mat separators soaked in electrolyte between the lead plates instead of liquid electrolyte.

AGM batteries are heavier and significantly more expensive than FLA but charge up to 3x faster than FLAs (without releasing dangerous gasses) [26]. They are also sealed (spillproof), require no maintenance, and can withstand more temperature and vibrations. Also, AGM self-discharge rate is just 1-3% per month (compared to 13% for FLA)[28].

### 7.1.3. LFP (lithium)

LFP cells are a type of Lithium-ion battery that use a Lithium Iron-Phosphate (LFP) cathode and a graphite electrode. LFP batteries have almost double the DoD of lead-acid batteries and can withstand many more cycles, commonly rated to 80% DoD for 4000 cycles. Furthermore, banks of LFPs are much more energy dense and for the same capacity can be **half** the weight of similar lead-acid options [29].

This comes at a cost however, as LFP batteries of a similar size to FLA or AGM can be 3-4x the price. Additionally, charging banks of LFP cells has its own challenges as Lithium-ion cells require specific charging profiles to maintain battery lifetime. This requires a BMS (Battery Management System) which increases electronic complexity and cost.

Table 8 Comparison of technical specifications of battery architectures considered. The closest battery capacities were chosen to compare the technology. The most desirable values for each characteristic are bolded.

| 12V Battery Comparison | Deep-cycle Flooded Lead-Acid | AGM Lead-Acid | Lithium-ion (LFP) |
|---|---|---|---|
| Model | Advanced L110 Non-Sealed Leisure Battery | Advanced AGM LPX110 Leisure Battery | Renogy 12V 100Ah Lithium Iron Phosphate |
| Price | **£100** | £150 | £450 |
| Capacity | 110 Ah | 110 Ah | 100 Ah |
| DoD (Depth of discharge – recommended max capacity use per cycle) | 50% | 50% | **80%** |
| Lifetime Cycles (at max DoD) | 400 | 500 | **2000** |
| Total Potential Lifetime Output (Watt·hours x DoD x lifetime cycles) | 264 kWh | 330 kWh | **1,920 kWh** |
| Cost per kW·hour | £0.38 | £0.45 | **£0.23** |
| Weight | 24 kg | 26 kg | **13 kg** |
| Temperature Range (discharging) | -15°C - 45°C | -15°C - 50°C | **-20°C - 60°C** |

## 7.2. DC-DC Converters

Power ratings of commercial equipment are compared in Table 9. 2.5kΩ is used as a typical ground path resistance from data supplied by equipment manufacturer ABEM.

Table 9 Commercial resistivity systems transmitter power specifications [32]

| Name | Max voltage | Max Power | Max current (at 2.5kΩ) |
|---|---|---|---|
| Syscal Junior | 400V | 100W | 160mA |
| ABEM SAS 1000 | 400V | 100W | 160mA |
| Syscal R1 Plus | 600V | 200W | 240mA |
| ABEM Terrameter LS2 | 600V | 250W | 240mA |

### 7.2.1. DC-AC Inverter

The first method considered is the most low-cost, simple method. A commercially available 12V DC-AC inverter connected to the battery generates 240V AC. Using a full-wave bridge rectifier and a smoothing capacitor 240VAC (RMS) is converted to 340VDC. The main benefit of this method is its extremely low-cost. This has been explored more in the CVR [2].

Technically, this is a crude approach, and the limited voltage means survey length, and therefore depth of investigation, is lowered.

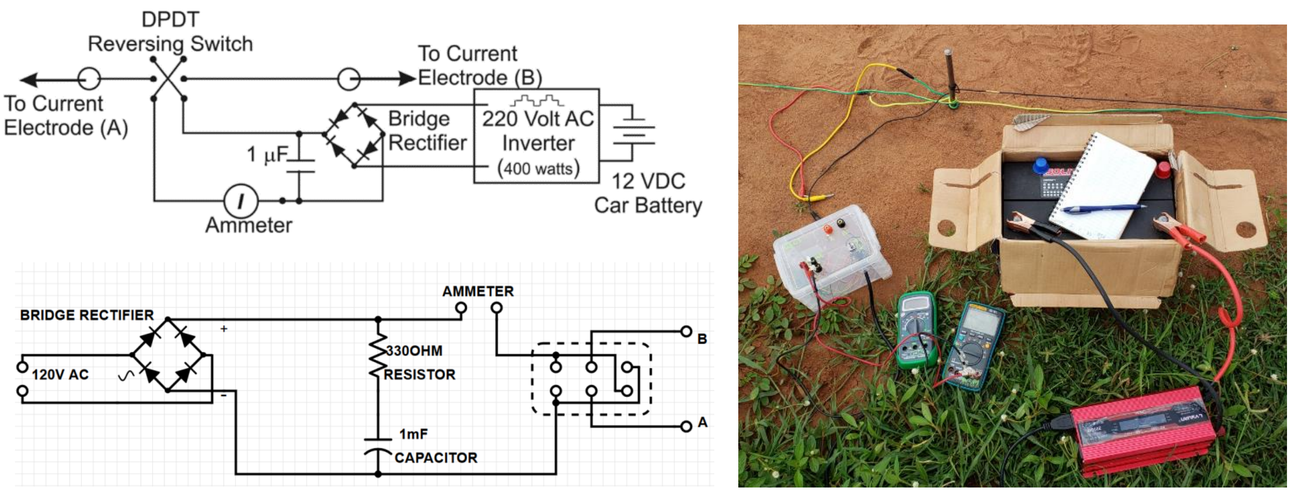



Figure 23 Left: Circuit diagrams from existing low-cost resistivity equipment studies (Top: Clark and Page (2011), Bottom: Sirota (2020)). Right: Sirota (2020) low-cost resistivity set up [34][35]

## 7.2.2. Flyback Converter

Flyback converters use a transformer, allowing higher or lower output voltages by varying the turns ratio. Figure 30 shows the method of operation. On closing the switch the primary of the transformer is energised – generating magnetic flux in the core, storing energy. Crucially, the secondary winding polarity is reversed. This negatively biases the output diode, so no current flows in the secondary while the primary is on. Opening the switch allows energy stored in the core to flow through the secondary, forward through the diode, charging the output capacitor and supplies the load.

Flyback converters are rated to <250W but can be designed with both a wide input range and extremely high output voltages. Another advantage of flybacks is their simplicity. Only 1 switch with 1 driver and no output inductors are required unlike other isolated topologies.

Figure 24 Examples of common household flyback converters (left: laptop charger; right: 12 VDC power supply)

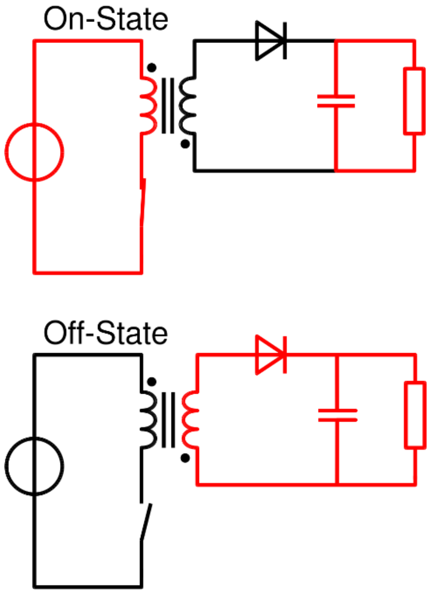



Figure 25 Flyback converter current paths based on switch state

# 8. Power Sub-System Discussion

## 8.1. Battery Selection

Standard FLA car batteries are functional, but not ideal in this context. Car starter batteries are optimised for high cranking amps to start an engine - not for deep-cycling or DoD past 20% repeatedly, which causes increased degradation of the plates. Deep-cycling batteries are therefore preferred.

Lithium-ion batteries are technically superior to lead-acid. However, LFPs extremely high price compared to lead-acid make them less appropriate in the context of the project aims. A lower cost deep-cycling lead acid battery is therefore more suitable.

The use case also does not justify using a more expensive AGM battery over a FLA. Flooded deep-cycle batteries cost less and can last a significant amount of time, providing suitable bang-for-buck while saving cost.

## 8.2. DC-DC Converter Selection

Using a DC-AC inverter as a power source is a crude method of generating the high voltages that has some proven effectiveness. However, it is limited by its low output voltage compared to commercial systems. For this purpose, the decision was made to develop a custom designed, open-source flyback converter that generates 500V+ from battery 12V, since flybacks can handle the required high gain and low power (<250W).

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# Appendices

## Appendix A Interactive Map Code

```python
locations = set()

# Create a map centered on South Africa
map_sa = folium.Map(location=[-25.513586, 28.098897], zoom_start=11)

# Add circle markers to the map for each unique location
for identifier in data['Identifier'].unique():
    # Get the data for this location and remove duplicate depth values
    location_data = data[data['Identifier'] == identifier].drop_duplicates(subset='DepthToBottom')

    # Filter out locations with no depth data
    if location_data['DepthToBottom'].isna().all():
        continue

    # Get the latitude and longitude of this location
    lat = location_data.iloc[0]['Latitude']
    lon = location_data.iloc[0]['Longitude']

    # Check if this location has already been plotted
    if (lat, lon) in locations:
        continue

    # Add this location to the set of plotted locations
    locations.add((lat, lon))

    # Create a circle marker for this location
    depth_labels = ['Layer {}'.format(i+1) for i in range(len(location_data))]
    table_rows = [[depth_labels[i], str(location_data.iloc[i]['DepthToBottom']), location_data.iloc[i]['LithologyName']] for i in range(len(location_data))]
    table_headers = ['Layer', 'Depth to bottom (m)', 'Lithology']
    tooltip_rows = ['<tr>{}</tr>'.format(''.join(['<td style="border: 1px solid black; padding: 5px;">{}</td>'.format(cell) for cell in row])) for row in table_rows]
    tooltip = '<table style="font-size: 14px; border-collapse: collapse;">{}<tbody>{}</tbody></table>'.format(
        '<tr>{}</tr>'.format(''.join(['<th style="border: 1px solid black; padding: 5px;">{}</th>'.format(header) for header in table_headers])),
        ''.join(tooltip_rows)
    )
    fill_color = 'red' if pd.isna(location_data['DepthToBottom']).all() else 'blue'
    folium.CircleMarker(
        location=[lat, lon],
        radius=3,
        fill_color=fill_color,
        color='blue',
        fill_opacity=1,
        tooltip=tooltip
    ).add_to(map_sa)

# Save the map
map_sa.save('borehole_map.html')
```

Borehole data within a 25km radius of Soshanguve available at: [Soshanguve boreholes 25km]

# Appendix B Kelvin Resistance Measurements

This measurement configuration in Figure 22 is called a **4-wire Kelvin measurement** since there are 4 leads attached to the target resistance, $R_{MN}$. Two force leads drive through the resistor, which creates a voltage drop according to Ohm's law (V=IR), and two voltage sense leads (**Kelvin connections**) are connected adjacent to the target resistance; eliminating the voltage drop in the force leads. Due to the high input impedance ($R_{input}$) of the voltmeter, almost no current flows through it and consequently there is negligible voltage drop in the voltage sense leads.

This is essential since parasitic resistances in 20m of cable can be more than 1Ω (rendering any measurement useless since $R_{MN}$ is comparably sized).

Figure 23 highlights the difference between 2 and 4-wire measurements. 2-wire measurements input current and measure resistance through the same 2 wires, inherently including the resistance of the measurement leads. 4-wire measurements remove the cable resistances, allowing pure measurement of the sample.

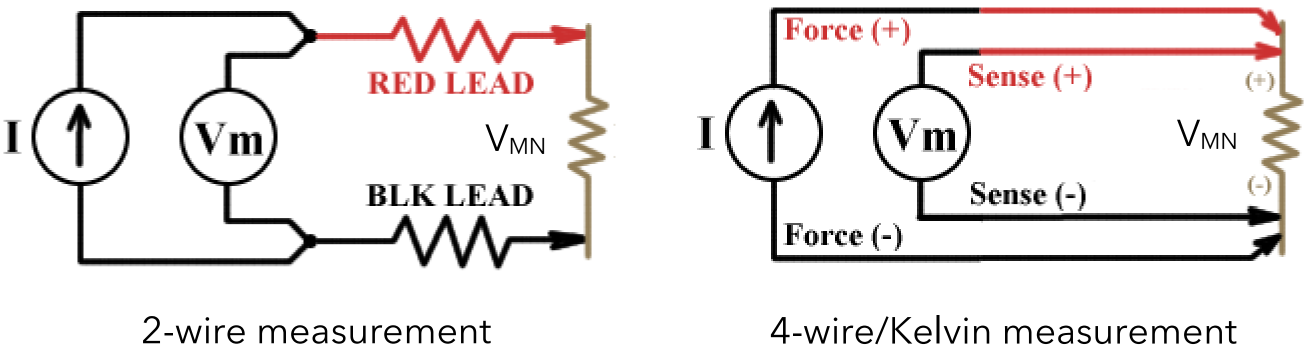



Figure 26 Diagrams of 2 (left) and 4-wire (right) sensing methods. The 4-wire diagram is a simplified version of the equivalent schematic in Figure 20

# Final Technical Design

## Power Electronics System

Candidate Number: **12245**

Group 1: Inaglobe

Word count: **2846**

# 1. System Overview

This report will focus on the detailed design and development of the power system for a DC resistivity system for groundwater exploration. Figure 28 shows the components and interfaces of the power supply sub-system. Ultimately, control of the power supply will be managed by an Arduino microcontroller. This will include sending signals to turn ON/OFF the DC-DC converter, battery charge monitoring, and output polarity switching via the H-bridge. Microcontroller design is out of scope of this report, which will focus on analysis and design of the DC-DC converter, and integration with the battery to form a complete power supply that meets all of the design requirements specified in the TFS.

Figure 27 represents the iterative design process followed during this project. From the initial requirements determined in the TFS, initial solutions were considered. This report focuses the implementation and evaluation of the solutions.

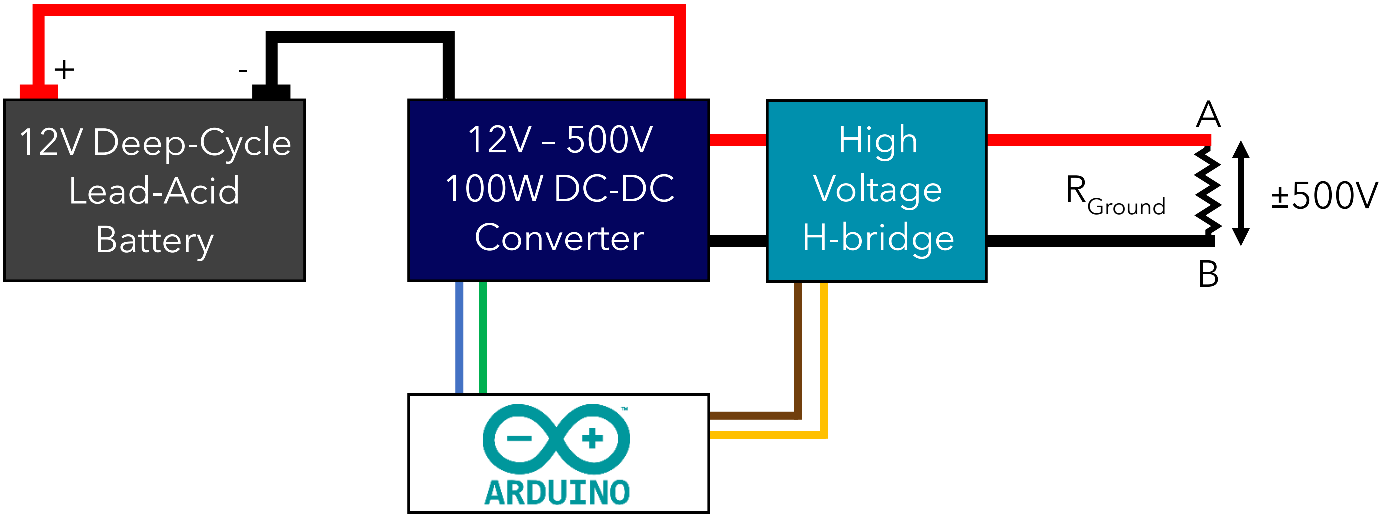



Figure 28 Power Supply Sub-system connections and interfaces

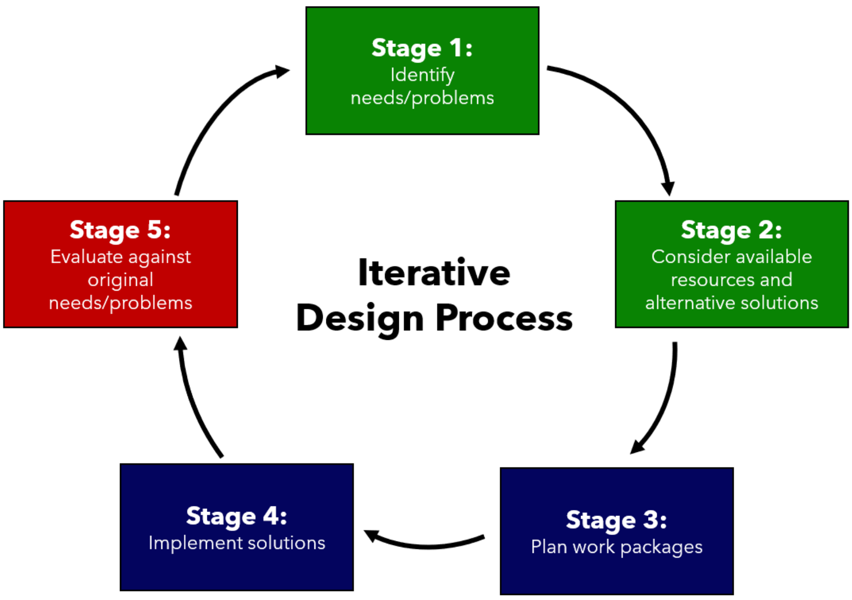



Figure 27 Representation of the design process followed during this project. Cyclic iteration and revisiting of the customer needs drives development by identifying gaps in solutions.

# 2. Concept Development

## 2.1. Flyback Converter Design

Flyback's use a 'coupled inductor' (a transformer with the secondary winding polarity reversed) in power conversion which galvanically isolates the input and output [1][2]. Figure 29 shows the general layout of an isolated flyback converter. At a very high frequency, a MOSFET switches the input voltage through the coupled inductor to drive the load through the output diode and capacitor.

Figure 30 shows the current paths taken in each switch state. While the MOSFET is on, current flows through the primary magnetising inductance. This generates magnetic flux in the core but does not energise the secondary since its polarity is switched, so forward current flow in the primary reverse biases the output diode. When the switch is off, the only route for magnetising inductance to go is backwards, flowing forwards through the diode and charging the output.

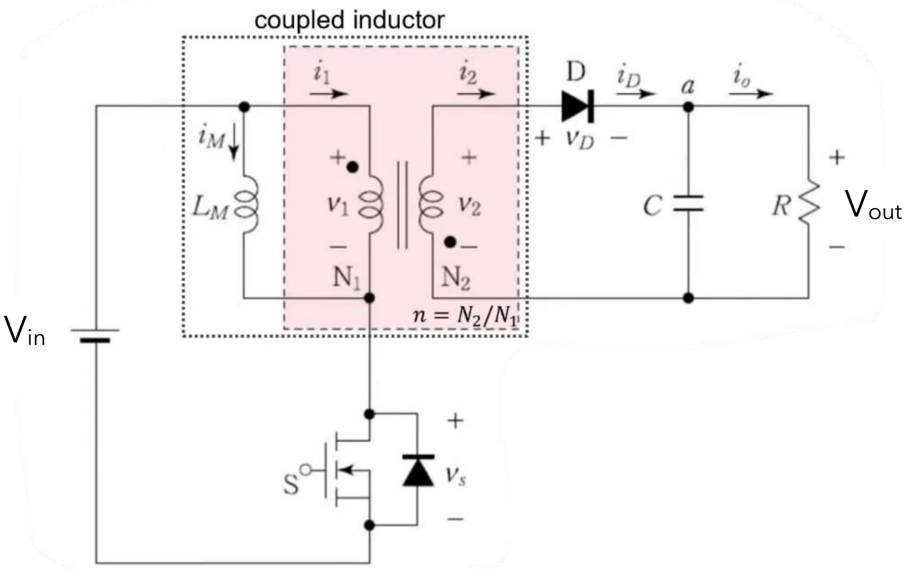



Figure 29 General layout schematic of a flyback converter

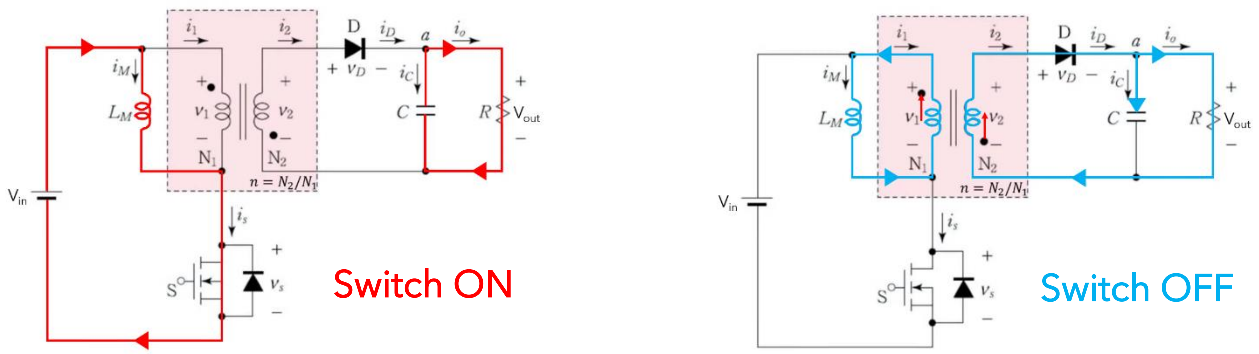



Figure 30 Current paths in a flyback converter for each MOSFET state [1]

## 2.1.1.Reverse Engineered Flyback

To begin to understand flyback converters the first step taken was to disassemble, analyse, and attempt to reverse engineer the schematic. The PCB in Figure 32 converts 240V AC mains into 30V DC using the method outlined above. The AC mains must first be rectified (converted to DC) before it is switched through the transformer.

This is not quite the same use case as this project which requires high step-up conversion but is nonetheless a useful design activity for understanding flyback converter design.

Figure 33 shows the reverse engineered schematic drawn in Falstad circuit simulator. It is not a 100% faithful reproduction, but it functions the same. A 555 timer IC has been used in place of a dedicated switching regulator to switch the MOSFET because it is the only option in Falstad.

### Takeaways from reverse engineering analysis

- Falstad is not an appropriate circuit simulation tool for high frequency switching power supplies (it is extremely laggy and slow for the speed and complexity of the circuit)
- 555 timers are not good as switching regulators, a dedicated switching regulator is preferred.
- High current traces can be tinned (i.e.: have extra solder added to increase trace thickness, decreasing resistance).
- A physical isolation gap between the high and low side of the converter is mandatory.
- Snubber circuitry can be used to dampen voltage ringing on the primary. This reduces the voltage stress on the MOSFET.
- Construction is quite basic, and flybacks are often made on single sided PCBs.

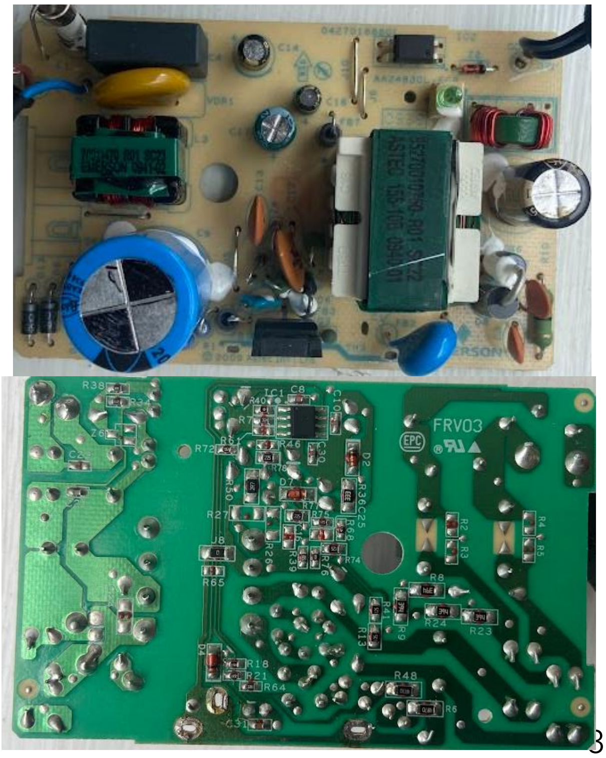



Figure 32 Flyback power supply for a printer. 240V AC mains to 30V DC

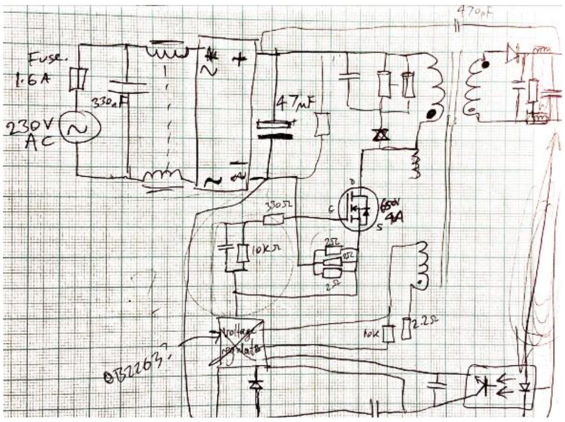



Figure 31 Hand drawn reverse engineered schematics quickly became messy.

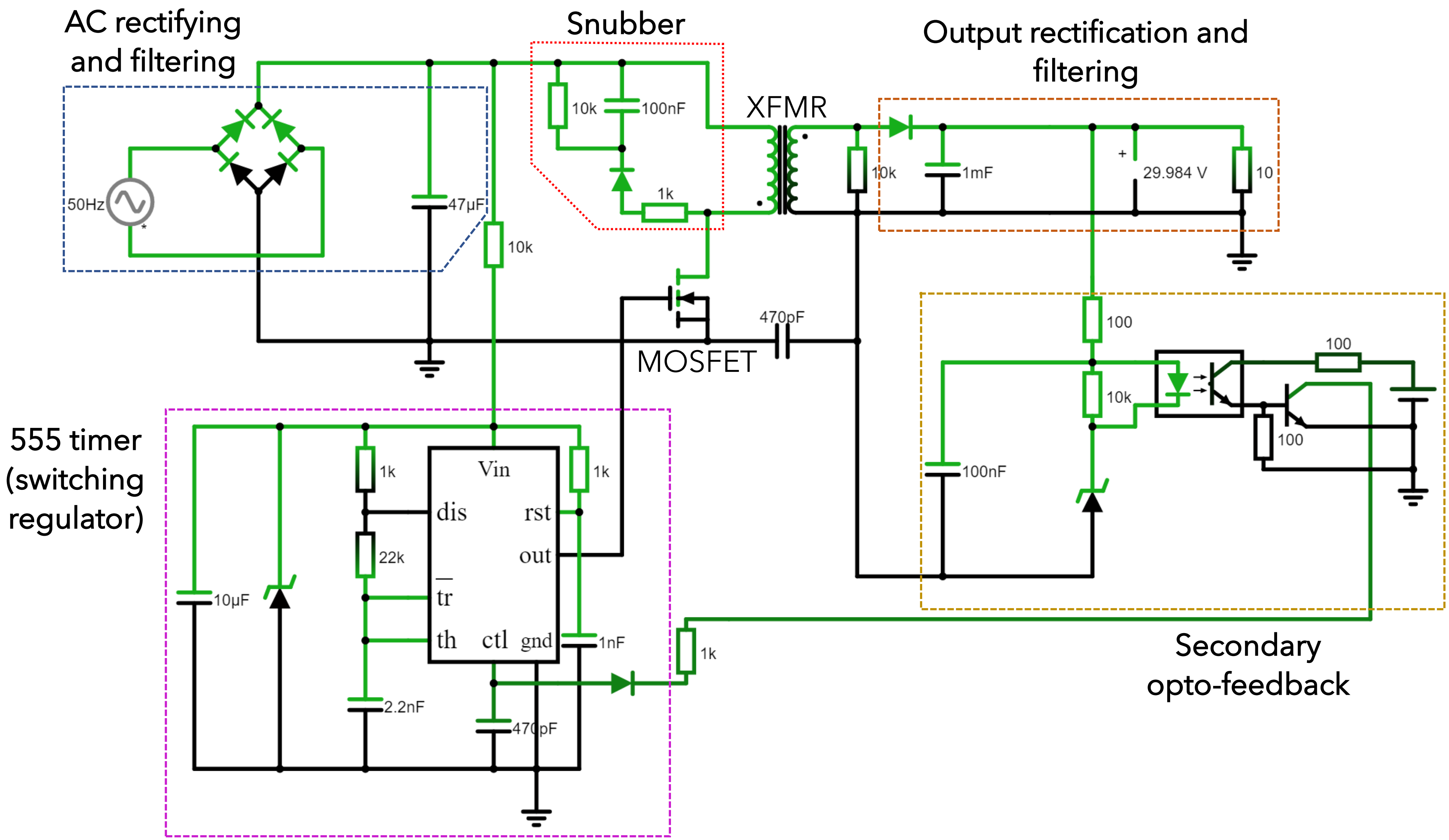



Figure 33 Labelled Falstad schematic of a flyback converter for 240V AC mains to 30V DC. A 555 timer is used as switching regulator with an optocoupler providing feedback from the output side to maintain isolation.

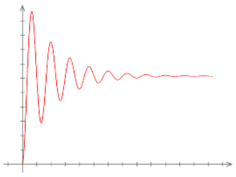
### 2.1.2. Switching Regulator

As a result of the previous analysis, choosing a dedicated switching regulator capable of meeting the requirements was required [3]. The main requirements of a switching regulator are:

100 kHz+

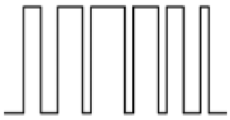
#### High Frequency Switching

The faster the switching frequency the smaller the transformer can be before it reaches saturation. Saving cost and weight.



#### Voltage Regulation

Monitoring the output voltage to reduce output ripple and respond quickly to step-changes in demand voltage.



#### Duty Cycle PWM

Modern ICs use active pulse-width modulation of the switch duty cycle to control the output voltage.

A range of semiconductor companies manufacture thousands of different ICs for flyback converters. Selecting a suitable regulator was done by parametrically searching manufacturer databases based on values Table 10, along with cost, and IC package (the physical shape of the chip and legs). See Appendix A for more details of the selection process.

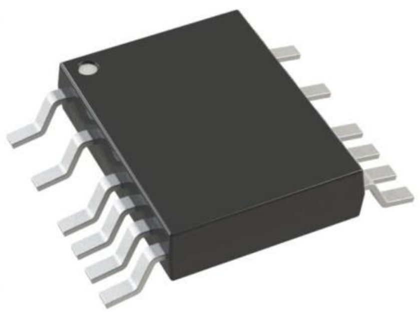
#### *2.1.2.1. Key Design Parameters*

Table 10 summarises design parameters defined by the power supply requirements specification in the TFS.

Table 10 Design parameters for the DC-DC converter power supply

| Design Input | Value |
|---|---|
| Input voltage ($V_{IN}$) | 10V to 13V |
| Output voltage ($V_{OUT}$) | 500V |
| Output current ($I_{OUT}$) | 200mA |
| Target efficiency ($\eta$) | 70% |

The LT3748 Controller from Analog Devices is specifically design for isolated flyback converters and capable of 100W+ power outputs [4]. The switching frequency is actively set based on load and primary inductance. Voltage regulation is done via 'primary side sensing', where the voltage across the primary winding is sampled during discharge (while the switch is OFF) to measure the output voltage. This eliminates the need for physical coupling between the primary and secondary sides – reducing part count, cost and complexity.



LT3748

### Specification

- Specifically designed for flyback converters
- High voltage Mini Small Outline Package (MSOP) saves PCB area
- Primary side sensing eliminates opto-coupler and auxiliary winding to regulate voltage
- 5V – 100V input range
- 100W+ output power
- Enable pin for easy power supply control
- Feedback pin allows dynamic output voltage control

## 2.1.3. Auxiliary Component Selection

The capabilities of the power supply are mainly limited by the sizing and specification of the auxiliary components (MOSFET, diode, capacitors, feedback resistors). Selecting appropriately rated parts will allow the DC-DC converter to fulfil the requirements.

### *2.1.3.1. Transformer Selection*

Selection of transformer turns ratio affects the whole design of the circuit and other requirements (MOSFET rating, Diode rating, output power, efficiency)[5]. Equations depending on this are:

$$N_{PS} = \frac{N_P}{N_S} \tag{1}$$

$$V_{DS} \geq V_{in_{max}} + (V_{out} + V_{F_{diode}}) \cdot N_{PS} \tag{2}$$

$$I_{out_{max}} \approx 0.85 \cdot (1 - D) \cdot N_{PS} \cdot \frac{I_{lim}}{2} \tag{3}$$

$$V_{R_{diode}} \geq \frac{V_{in_{max}}}{N_{PS}} + V_{out} \tag{4}$$

Equation 1 is the turns ratio of primary:secondary ($N_{PS}$). Equation 2 describes the flyback voltage reflected onto the primary. This is the mechanism used by the regulator to monitor output voltage described above, but it also defines the voltage rating of the MOSFET. $V_{DS}$ is the MOSFET drain-source voltage, the minimum voltage it must withstand while off. This requirement depends on the input voltage ($V_{in}$), output voltage ($V_{out}$), forward drop of the diode ($V_{F,diode}$) and the turns ratio [4].

Equation 3 defines the output current (and power) of the converter. This relies on the turns ratio, duty cycle (D), the primary-side current limit ($I_{Lim}$), and assumes an efficiency of 85%. Equation 4 sets the reverse bias rating of the output rectification diode ($V_{R,diode}$). As shown in FIG3, while the MOSFET is on this must withstand the full primary and secondary voltage in reverse without breaking down.

Once an appropriate turns ratio has been selected the transformer inductance limits can be defined:

$$L_{pri} \leq \frac{V_{in} \cdot \left(V_{out} + V_{F_{diode}}\right) \cdot N_{PS}}{f_{sw_{min}} \cdot I_{lim}[(V_{out} + V_{F_{diode}}) \cdot N_{PS}] + V_{in_{min}}} \tag{5}$$

$$L_{pri} \geq \frac{\left(V_{out} + V_{F_{diode}}\right) \cdot N_{PS} \cdot R_{sense} \cdot 400\text{ ns}}{15\text{ mV}} \tag{6}$$

$$L_{pri} \geq V_{in_{max}} \cdot R_{sense} \frac{250\text{ ns}}{15\text{ mV}} \tag{7}$$

Bounding values for transformer inductance are set by internal circuit constraints of the LT3748 [4]. Equations 6 and 7 set minimum inductances based on the minimum time required for the IC to sample the flyback voltage and minimum time the gate node can be set high, respectively. Equation 5 is the upper bound and is based on the minimum switching frequency ($f_{sw,min}$) at maximum load.

Equations 1-7 have been evaluated together in the spreadsheet shown in Table 11. This allows comparison of different turns ratios and the effect of changing input parameters on component requirements.

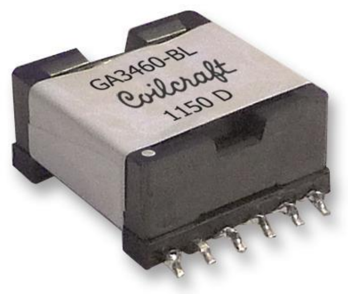
Table 11 highlights the trade-offs present in selecting a transformer. Decreasing the turns ratio puts less stress on the MOSFET ($V_{DS}$), but increases the diode reverse rating ($V_{R,diode}$). A higher turns ratio draws less current through the diode ($I_{diode,rms}$) which will increase efficiency, but this greatly increases the required primary inductance ($L_{pri2}$).

From this analysis the most appropriate turns ratio was evaluated as 1:10. This is the highest turns ratio that meets inductance requirements, while minimising diode RMS current.

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $V_{in}$ 12 | | $V_{in,min}$ 11 | | | $V_{out}$ 500 | | | $V_{f(diode)}$ 1.15 | | | | |
| | | $V_{in,max}$ 13 | | | $I_{out(max)}$ 0.2 | | | $f_{sw(min)}$ 80000 | | | | |
| | | $V_{in,nom}$ 12 | | | | | | | | | minimum inductances: | |
| | | | | | | $V_{in,min}$ | $V_{in,nom}$ | | | max | $t_{settle}$ | $t_{on}$ |
| pri:sec | $N_{ps}$ | $V_{ds}$ | $V_{r(diode)}$ | D (Vin = 11V) | D (Vin=12V) | I(lim) (50mA @ Vin = 11V) | I(lim) (50mA @ Vin = 12V) | Idiode(rms) (Vin = 12V) | $R_{sense}$ (Ω) | $L_{pri1}$ < µH | $L_{pri2}$ > µH | $L_{pri3}$ > µH |
| 1:50 -> | 0.02 | 23 | 1150 | 0.48 | 0.46 | 47.78 | 45.88 | 19.56 | 0.0022 | 3.00 | 0.58 | 0.47 |
| 1:20 -> | 0.05 | 38 | 760 | 0.69 | 0.68 | 32.78 | 30.88 | 10.16 | 0.0032 | 4.45 | 2.16 | 0.70 |
| 1:10 -> | 0.1 | 63 | 630 | 0.82 | 0.81 | 27.78 | 25.88 | 6.59 | 0.0039 | 5.31 | 5.16 | 0.84 |
| 1:5 -> | 0.2 | 113 | 565 | 0.90 | 0.89 | 25.28 | 23.38 | 4.45 | 0.0043 | 5.88 | 11.43 | 0.93 |
| 1:4 -> | 0.25 | 138 | 552 | 0.92 | 0.91 | 24.78 | 22.88 | 3.95 | 0.0044 | 6.01 | 14.60 | 0.95 |
| 1:3 -> | 0.33 | 180 | 539 | 0.94 | 0.93 | 24.28 | 22.38 | 3.40 | 0.0045 | 6.14 | 19.90 | 0.97 |
| 1:2 -> | 0.5 | 264 | 526 | 0.96 | 0.95 | 23.78 | 21.88 | 2.76 | 0.0046 | 6.28 | 30.54 | 0.99 |
| 1:1 -> | 1 | 514 | 513 | 0.98 | 0.98 | 23.28 | 21.38 | 1.98 | 0.0047 | 6.43 | 62.50 | 1.01 |

Table 11 Iterative comparison of a range of turns ratio values to quantify the required transformer inductance, diode reverse rating and MOSFET drain-source voltage

Practically, there is a very limited selection of "stock" transformers that fit the requirements. This is because most power supplies are manufactured in such bulk that custom transformers can be manufactured directly to designers specifications [6]. Further explored in the Commercial Viability Report, there is not enough demand for the product to justify a custom transformer, so one of the limited stock transformers close to the specification must be chosen.




## Specification [7]

- 1:10 turns ratio
- 500V maximum output
- 5V – 24V input
- 50A current limit
- 5μH primary inductance
- Flux shield to minimise EMI emissions
- Specifically designed for flyback converters
- Designed for 'capacitor charging' circuits with similar specifications to this study

### *2.1.3.2. MOSFET Selection*

Column 3 in Table 11 sets the MOSFET drain-source voltage. This is the voltage it must withstand while the switch is open, set by equation 2. For a 1:10 transformer this is nominally 63V. However, for a non-ideal transformer, leakage inductance (imperfect magnetic coupling between the primary and secondary) can double this value.

Similarly to regulators, there are thousands of MOSFETs available which meet the requirements [8]. Filtering and selection can be done using a Figure of Merit (equation 8) metric. This is the product of two internal characteristics of the MOSFET: $R_{DS,on}$ the drain-source resistance when the MOSFET is on, and $Q_G$ the MOSFET gate capacitance. The lower the FoM the better [9].

$$\mathrm{FoM} = \mathrm{R_{DS_{on}}} \cdot \mathrm{Q_G} \tag{8}$$

Similarly to the regulator, parametric filtering and selection using FoM metrics was undertaken (details in Appendix B). The final MOSFET selection is the **Infineon IPP200N15N3**.


## Specification [10]

- 150V drain-Source Voltage ($V_{DS}$)
- 20mΩ on-resistance ($R_{DS,on}$)
- 50A maximum drain current ($I_D$)
- 23nC gate capacitance ($Q_G$)

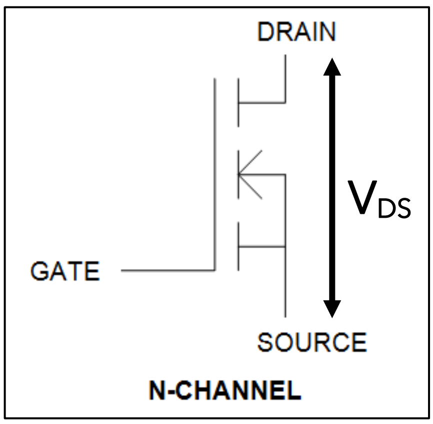



Figure 34 Drain-Source voltage is the maximum voltage that can be applied across the MOSFET while it is not conducting

$$I_{FET_{RMS}} = \sqrt{I_{Lim}^2 \cdot \frac{D}{3}} = \sqrt{27.8^2 \cdot \frac{0.82}{3}} = 14.53\ \text{A} \tag{9}$$

$$P_{FET} = {I_{FET_{RMS}}}^2 \cdot R_{DS_{on}} = 14.53^2 \cdot 0.02 = \mathbf{4.22W} \tag{10}$$

Equations 9 and 10 [4] calculate an estimated 4.22W are dissipated in the MOSFET using values in Table 11. This is a ~5% loss of efficiency, mainly due to the high primary side current required by the use of a 1:10 transformer.

#### *2.1.3.3. Diode Selection*

The output diode reverse voltage is the maximum voltage that can be applied backwards across the diode without allowing current flow. Table 11 specifies $V_{R,diode}$ = 630V, so the 800V rated **ROHM RFN5TF8SC9** [11] was selected. This is an 'ultra-fast' diode designed for rectification in SMPSs and can switch direction in 40ns (reverse recovery).

#### *2.1.3.4. Passive Components*

The LT3748 also requires specific resistors to set the output voltage and power.

$R_{sense}$ is a current sense resistor used to set the primary current limit. Equation 11 sets this value - Table 11 shows this should be 4mΩ, although this is an uncommon value so **5mΩ** was chosen.

$$R_{sense} = \frac{100\ \text{mV}}{I_{lim}} \tag{11}$$

$R_{FB}$ is the feedback resistor used to program the output voltage. This is calculated in equations 12 and 13:

$$R_{FB} = \frac{R_{ref} \cdot N_{ps}\left[\left(V_{out} + V_{F_{diode}}\right) + V_{TC}\right]}{V_{BG}} \tag{12}$$

$$R_{FB} = \frac{6040 \cdot 0.1[(500+1.6)+0.55]}{1.223} = 248\text{k}\Omega \tag{13}$$

$R_{ref}$, $V_{BG}$ and $V_{TC}$ are internal values set by the LT3748. A more standard value of **250kΩ** was chosen to simplify part procurement.

### 2.1.4. Schematic Simulation Iterations

The industry standard simulation tool for modelling is switching power supplies is LTspice [12]. This makes using the LT3748 regulator convenient as it is built into the program, so the expected behaviour of the power supply can be accurately modelled. Iteration and testing of the circuit in a SPICE simulation allows quick, free insights into circuit behaviour without purchasing components or handling the potentially dangerous voltages involved.

Adapting examples from the LT3748 datasheet [4] and using calculations in Section 2.1.3, a first schematic iteration was produced in LTspice (Figure 35). The values in this will then be tuned in the following sections to better understand and refine the behaviour.

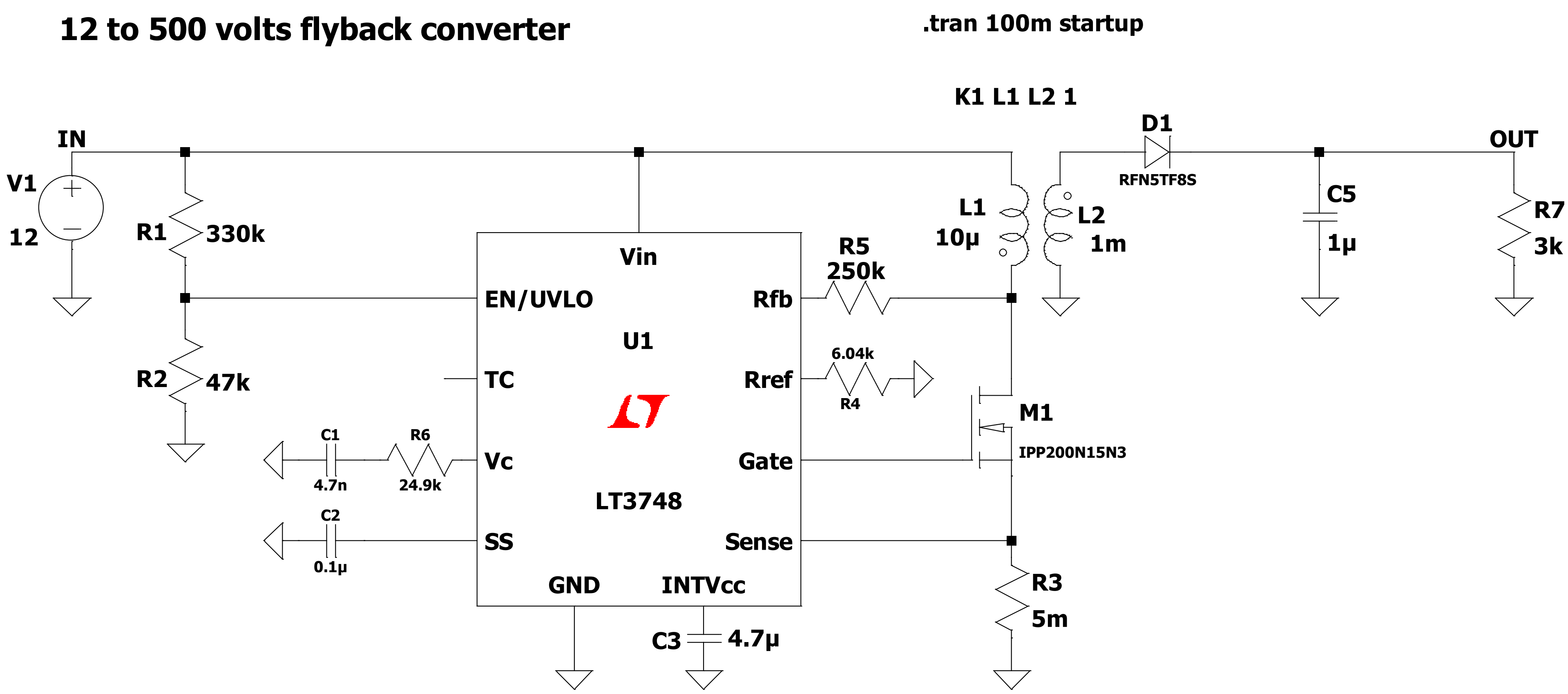



Figure 35 First iteration of DC-DC flyback converter schematic

Figure 36 shows the transient start-up performance of the schematic. A 3kΩ load is used as an approximation for common ground path resistances (typical values in Appendix D). In this configuration the DC-DC converter generates 530V across the load, driving 175mA, for an output power of **93W**.

This is sufficiently close to the target output parameters and fulfils the requirements of the DC-DC converter. However, iteration and improvement can be done to refine and further characterise the performance of the converter.

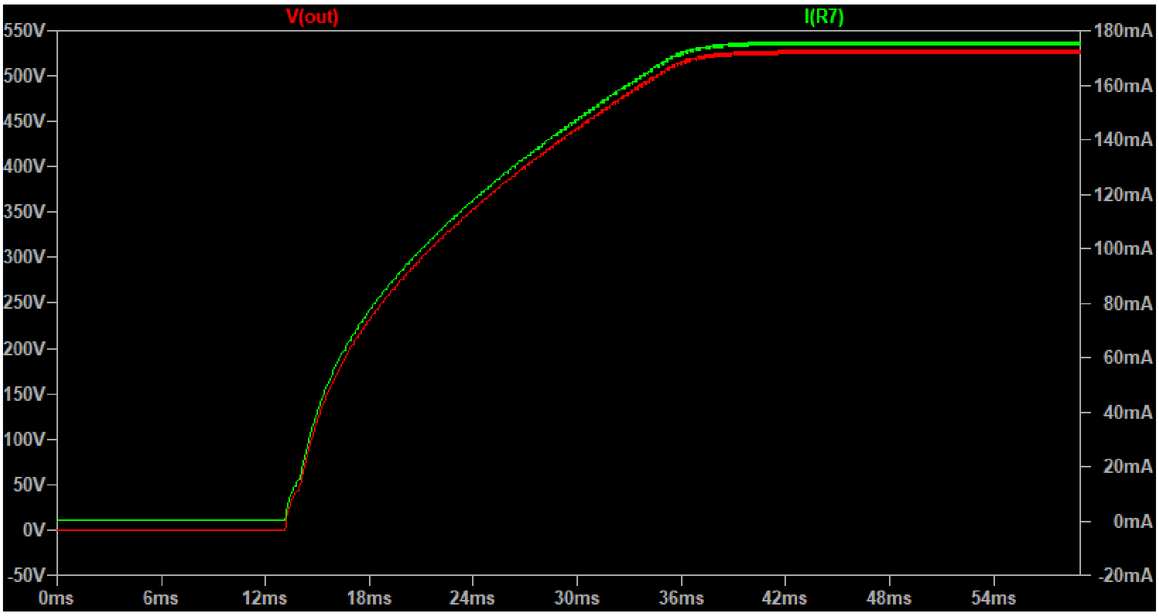



Figure 36 Start-up transient performance of the DC-DC converter first iteration

### *2.1.4.1. Output Filtering Capacitors*

Smoothing the output voltage can be done by adding filtering capacitors. The output ripple ($\Delta V_{max}$) can be calculated with equation 14 [4]:

$$\Delta V_{max} = \frac{L_P \cdot I_{Lim}^2}{2 \cdot C_{out} \cdot V_{out}} \tag{14}$$

Figure 37 shows the initial 1μF capacitor results in almost 5V (1%) ripple.

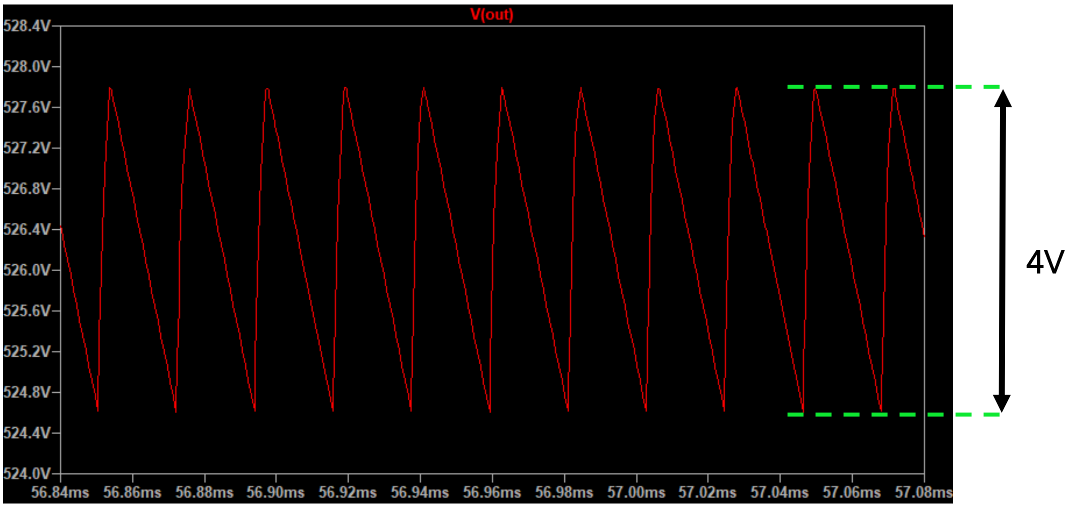



Figure 37 Output ripple using initial selection of a 1μF capacitor

Rearranging for output capacitance, a desired maximum ripple can be inputted, in this example $\Delta V_{max} = 1V$ is used.

$$C_{out} = \frac{L_P \cdot I_{Lim}^2}{2 \cdot \Delta V_{max} \cdot V_{out}} = \frac{5\mu H \cdot 27.8A^2}{2 \cdot 1V \cdot 530V} = \mathbf{3.6\mu F} \tag{15}$$

The effective of varying output capacitance is demonstrated in Figures. Increasing this capacitor to 3.6μF reduces the ripple to 0.9V in line with equation 15, but the power supply takes almost twice as long to reach steady-state operation.

The output capacitor could be increased more to smooth the output further. This is not necessary because the system is not sensitive to such small high-frequency ripple. 0.2% voltage ripple translates to 0.3mA of current ripple which will not affect measurements noticeably. Larger capacitors also incur more cost, so it is best to size them appropriately.

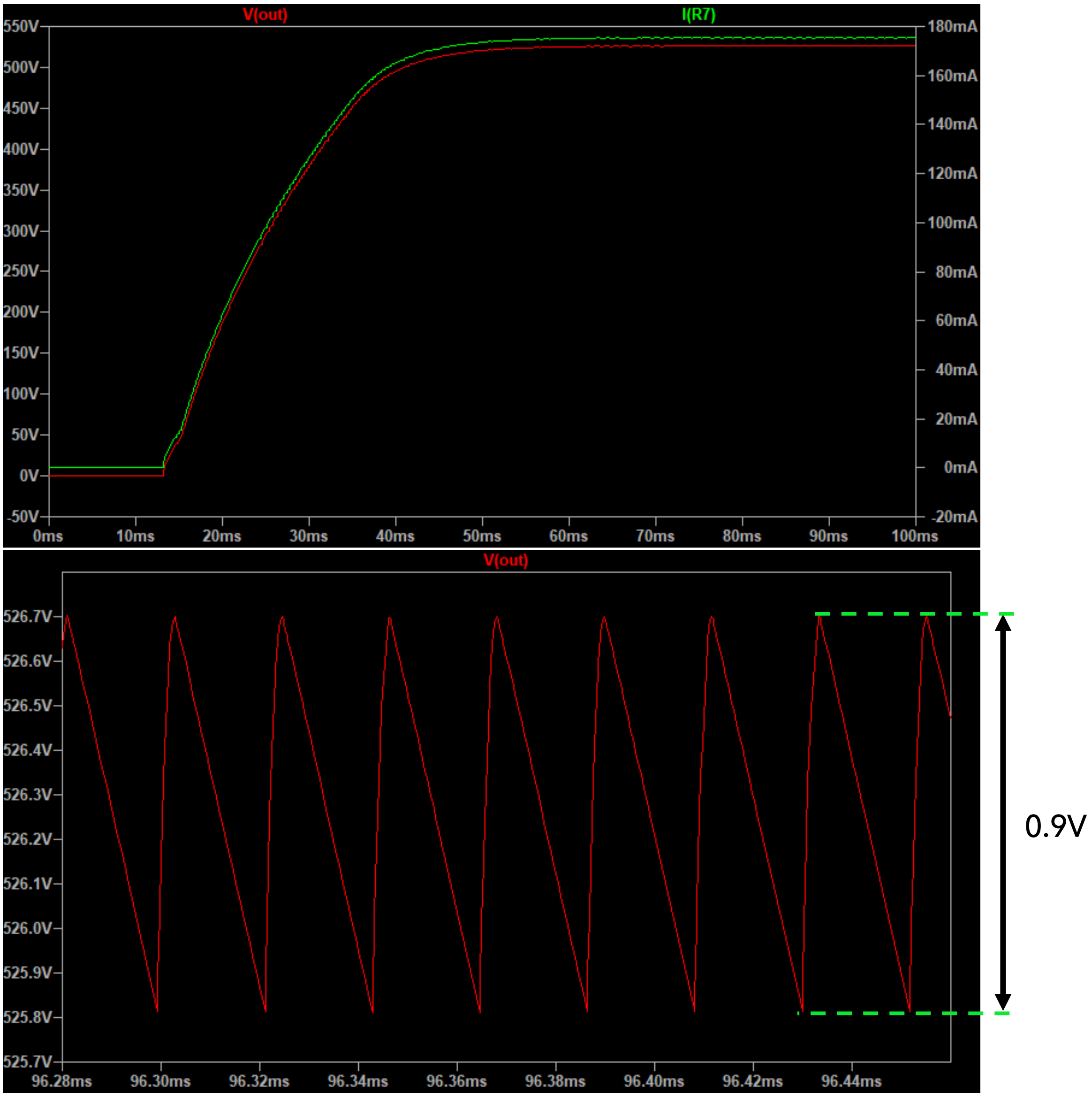



Figure 38 Top: transient start-up using larger filtering capacitors. Bottom: Output ripple reduction to 0.9V using larger capacitors

### 2.1.4.2. *Feedback Resistor Variations*

$R_{FB}$ (R5 in Figure 35), the feedback resistor connected to node Rfb on the LT3748 programs the output voltage by detecting the reflected voltage from the primary to secondary during discharge. Varying this resistor theoretically could allow dynamic control of the output voltage according to equation 12.

This has been tested in LTspice to confirm and characterise this behaviour (Figure 39). Figure 40 shows a comparison of the theoretical dependency of $V_{out}$ on $R_{FB}$ according to equation 12, with the simulated values generated in LTspice. Figure 39 and Figure 40 confirm the possibility of dynamically controlling the output voltage.

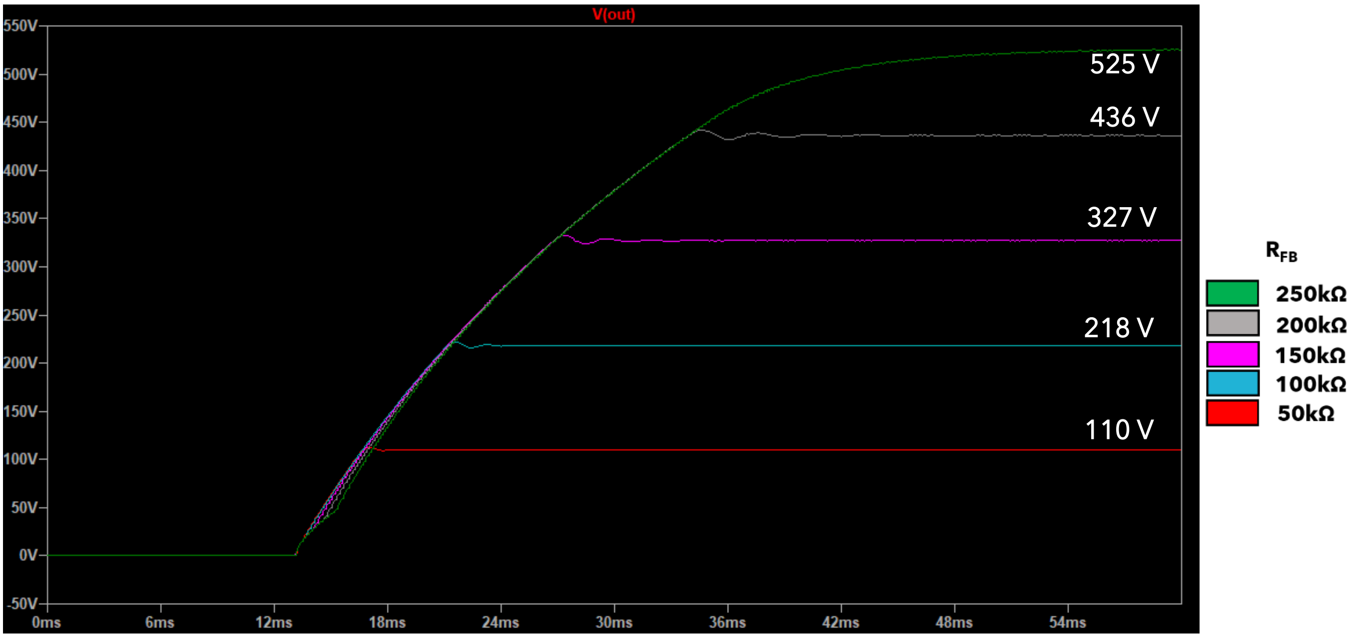



Figure 39 LTspice simulation results for varying values of $R_{FB}$

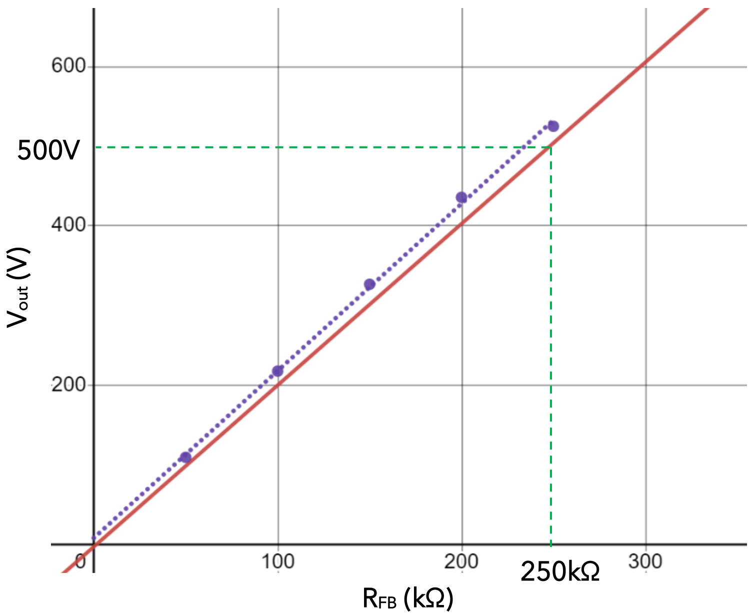



Figure 40 Variation of output voltage ($V_{out}$) with feedback resistance value ($R_{FB}$). Theoretical relationship (red) in comparison with simulated behaviour (purple). Original selection of 250kΩ for 500V is shown (green)

### 2.1.4.3. *Sense Resistor Variations*

Figure 41 shows the effect of varying the $R_{sense}$ resistor (R3 in Figure 35). The current is 'sensed' by the precise low resistance as it generates a small voltage drop across it. Using Ohm's law, the current flow can be calculated. Decreasing $R_{sense}$ reduces the voltage across it , allowing more current to flow and therefore more output power.

However there are practical problems with very low (<5mΩ) $R_{sense}$ values. At this low resistance the connecting PCB traces can have a significant parasitic resistance – increasing the effective resistance between the LT3748 sense pin and ground; reducing the output voltage.

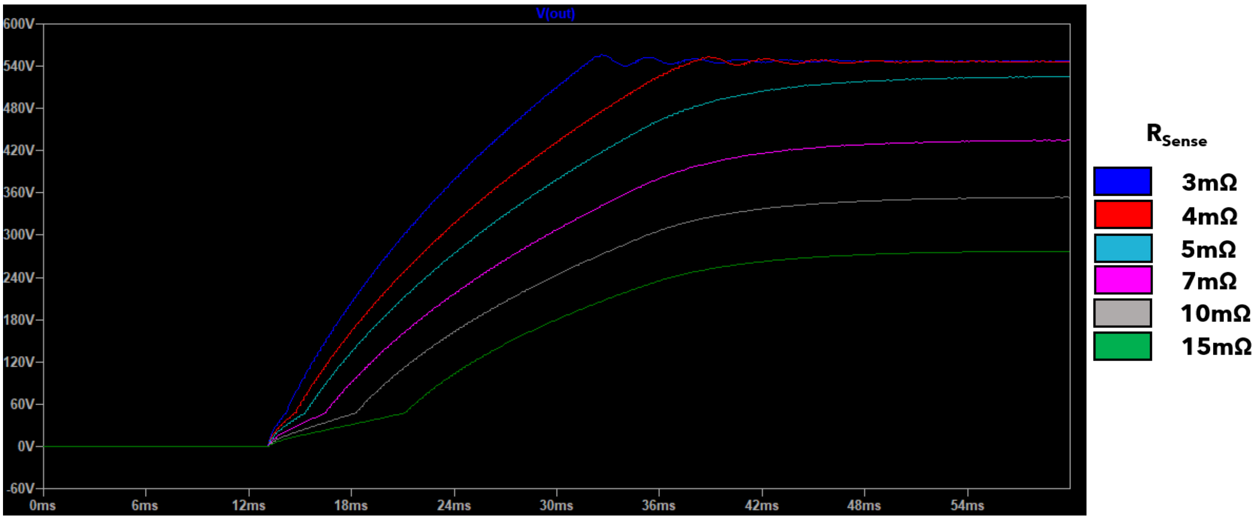



Figure 41 Effective of varying $R_{sense}$ on output voltage. Practically however, it is difficult to reduce $R_{sense}$ below 5mΩ

## 2.2. Battery Power

### 2.2.1. Battery Selection

Several battery architectures have been compared and contrasted the TFS. Ultimately, the best solution considering commercial and technical aspects was a deep-cycling flooded lead-acid battery. The key requirements of the battery are:

**Deep-Cycling**

Discharging to 50% or more allows remote multi-day surveys without damaging the battery due to low charge.

**Low-cost**

Cheaper battery technology that meets the power requirements is preferred to improve access to the system.

Commercial
Off
The
Shelf

**COTS**

A fully off-the-shelf solution is preferred to avoid incurring unnecessary system complexity and cost

Power requirements and capacity have been calculated in the TFS. However, deep-cycling lead-acid batteries are most commonly made for leisure purposes (caravans/motorhomes), so the smallest sizes commonly available are around 85Ah. A heavy day of surveying may only use 6.6Ah. For a 50% DoD this size battery would only need charging once a week.


**Specification [13]**

- Deep-cycle depth of discharge (DoD) 50%
- Sealed Flooded Lead-Acid
- 12 V
- 85 Ah capacity
- 21 kg
- 400 cycles
- 2 year warranty

Figure 42 (Left) Heavy duty jump leads with Anderson connector [14]. (Right) 12V battery charger [15]

## 2.2.2. Auxiliary Parts

A battery charger and connector cable are also required. Table 11 shows that the maximum input current is 27A. Figure 42 shows two COTS parts that fulfil the requirements.

The jumper cable uses an Anderson plug (a heavy-duty secure connection for high currents) to interface between the battery and the power converter and is rated for 240A. It cannot be plugged-in the wrong way which adds additional protection to the internal circuitry.

COTS battery chargers are widely available, and an example is included here for costing purposes. The customer would be expected to source their own charger and battery (due to complications and costs when shipping batteries).

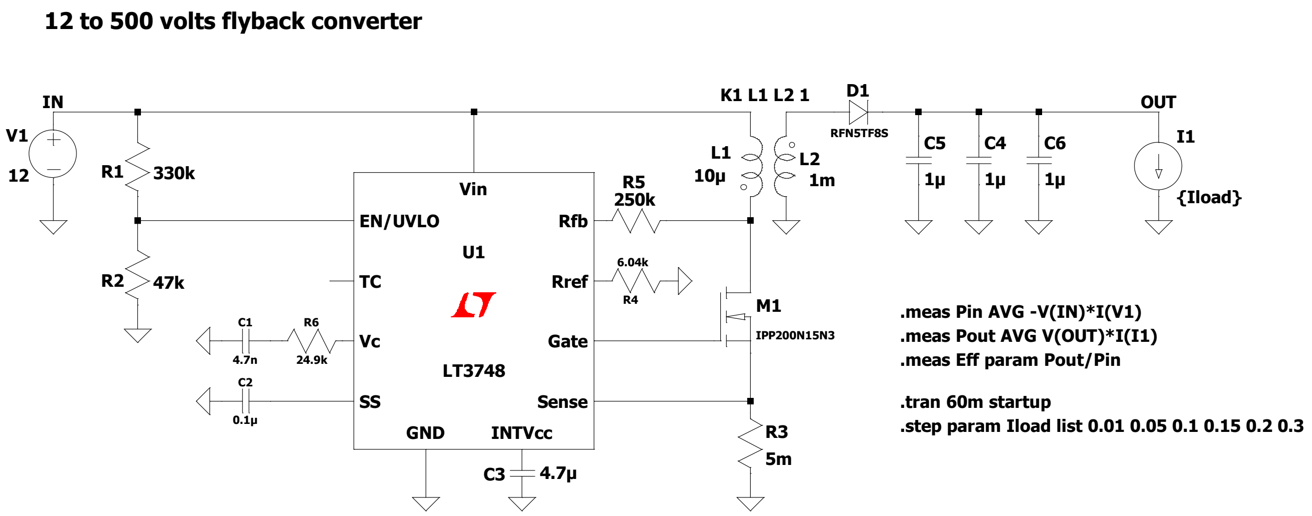



Figure 43 Final DC-DC flyback converter schematic

## 2.3. Final Schematic and Evaluation

Figure 43 shows the final converter schematic after incorporating the changes and considerations in the previous section. A complete BOM of the converter is in Appendix E. To demonstrate functionality, the converter start-up transients are shown in Figure 44. These have been run over a number of output loads to mimic real earth resistances experienced in the field and characterise the expected performance of the power supply.

Typical earth path resistances are usually between 1.5kΩ – 3kΩ (Appendix D). Current loads of 10mA to 300mA have been used to represent Earth resistances of 1kΩ to over 50kΩ. Conductive clay-rich soil have the potential to overload the power supply by drawing too much current as demonstrated below [16].

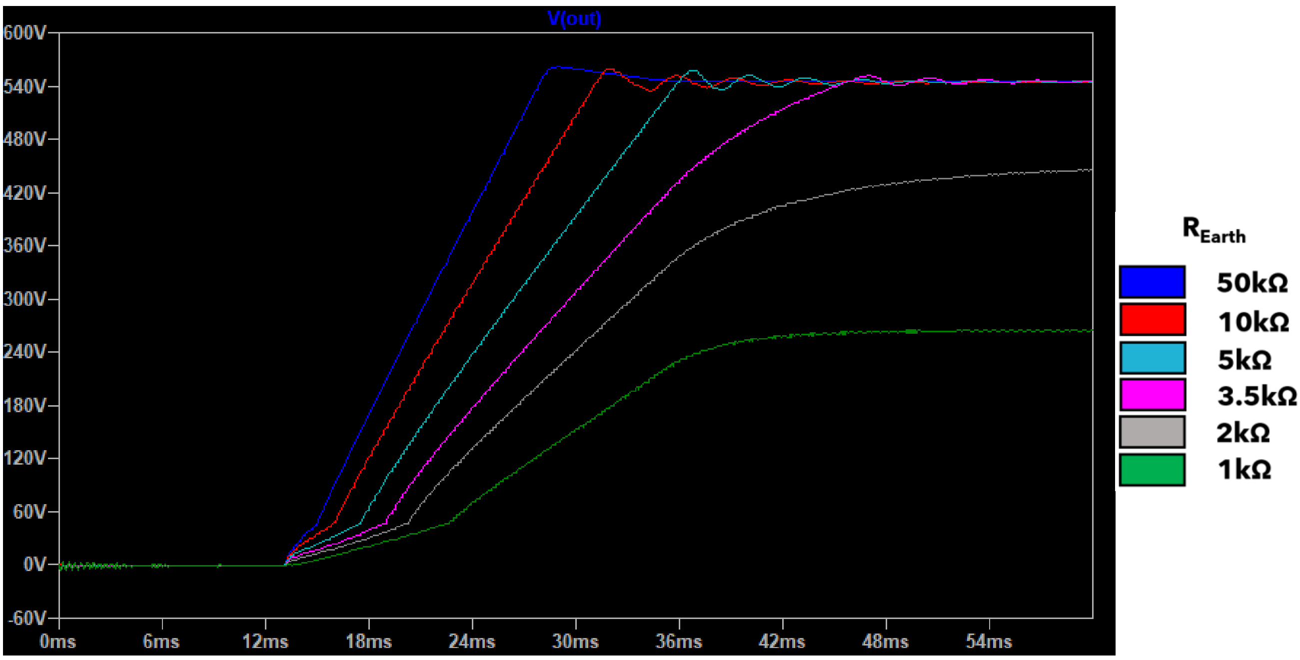



Figure 44 Output voltage over a range of Earth resistances to estimate the performance of the power supply in real world conditions

### 2.3.1. Efficiency Calculations

Figure 43 also includes modelling parameters to calculate the efficiency of the converter at each output level [17]. This is done while the output is in steady-state by comparing the input voltage (V1) to the output load (I1). Efficiency over a range of load currents is shown in Figure 45. This highlights the ineffectiveness of the power supply in high conductive soils, where its power output is not enough to maintain the high voltage.

Efficiency is not a crucial characteristic of the converter however, as it is only on for a few seconds every few minutes while data is being recorded.

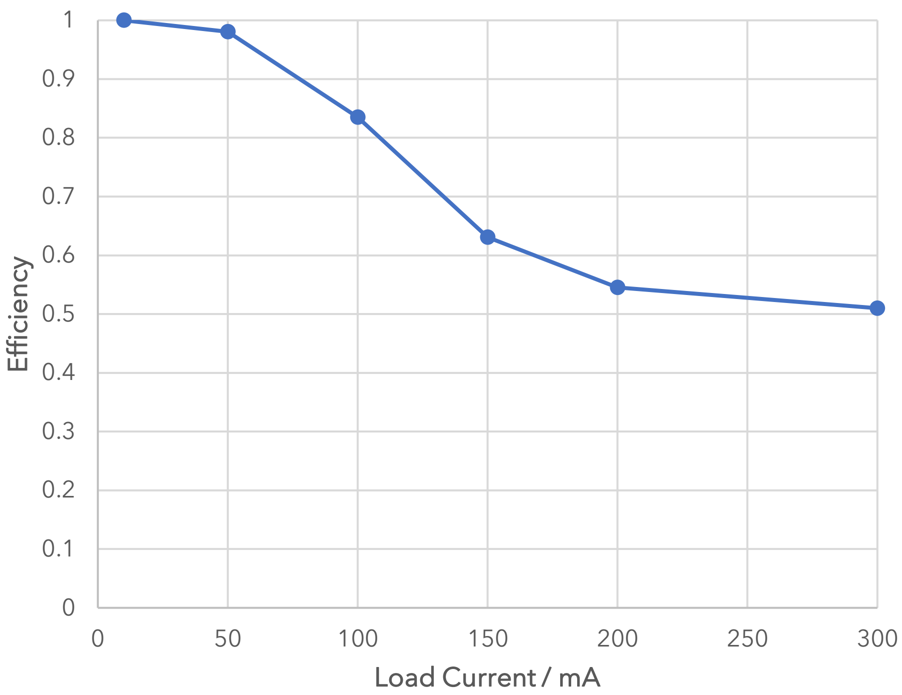



Figure 45 Converter efficiency at a range of current loads, corresponding to decreasing earth resistance showing how the converter becomes overloaded

### 2.3.2. Solution Lifecycle Considerations

The overarching theme of this project is increasing groundwater access by making geophysical equipment more accessible. To this end, the ultimate goal is to make the design open-source and draw on large online communities of engineers to improve designs and support people.

However, another route is to manufacture the system on a small-scale (100's) ,to generate profit for the partner NGO, creating a sustainable business asset rather than contributing to a collective open-source knowledge-base.

These two options are broken down in Figure 46 and discussed further in the Commercial Viability Report.

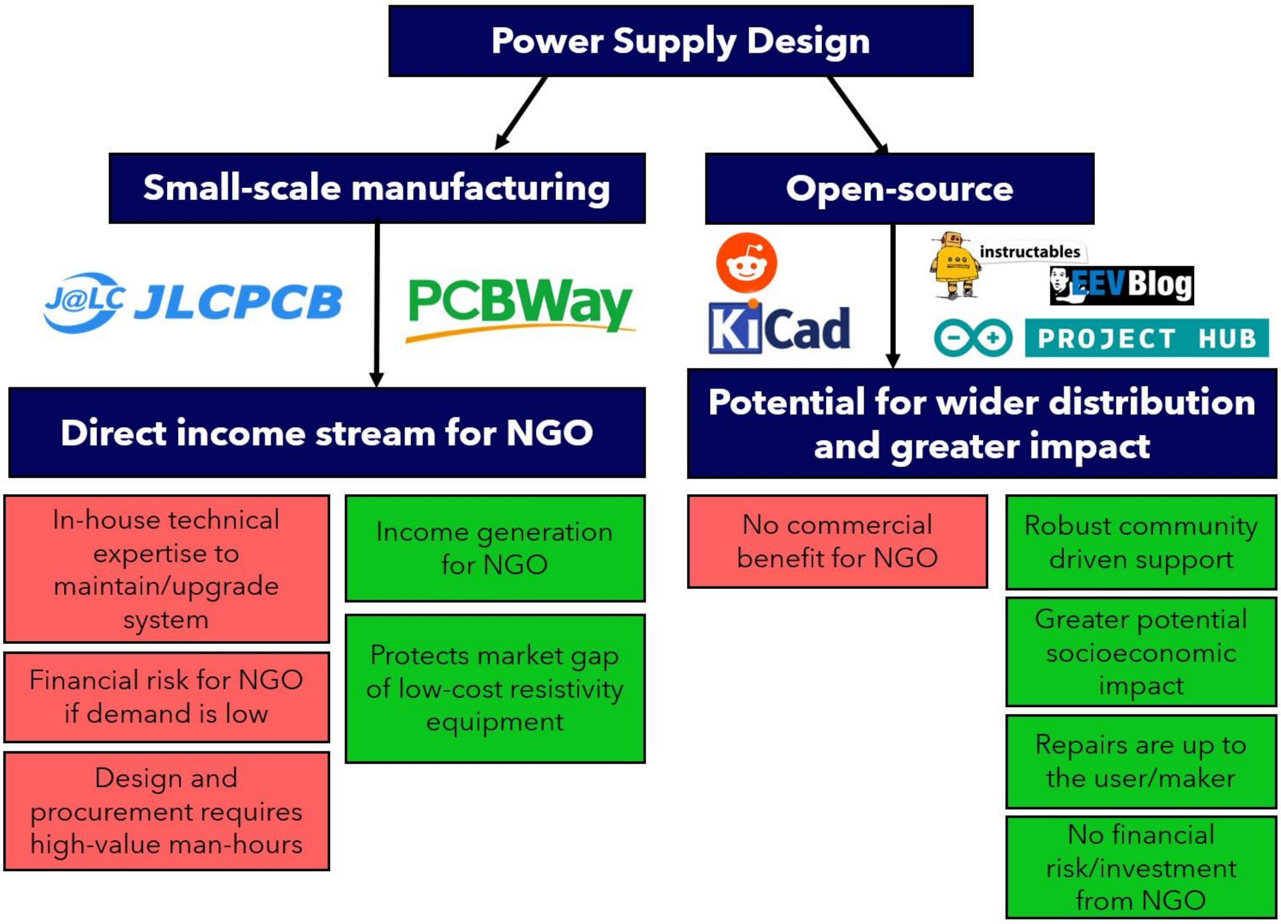



Figure 46 Summary of considerations for the life cycle routes of the power supply design. This topic is discussed further in the Commercial Viability Report for this project

Protecting the design and manufacturing the power supply for profit has potential to bring more income to the NGO and create a sustainable business around the product. This could be done using an accessible PCB assembly service like PCBWay or JLCPCB, with cost estimates around £2000 for 200 assembled PCBs [18]. However, protecting the designs has financial risks if they cannot sell the units, and upgrading/support for sold systems requires in-house technical expertise which the NGO may not be able to afford.

Open-sourcing the design has the potential to have a much greater socioeconomic impact by utilising the worldwide maker community to drive support and further development of the project.

# 3. Project Review

## 3.1. Solution Specification

Evaluation of the final design against the power supply requirements specification is summarised below (Table 12).

Table 12 Evaluation of the final Power System specification against the design requirements specification

| Requirement | Target | Must/ Wish | Achieved? | Comment |
|---|---|---|---|---|
| **Battery** | | | | |
| Battery can power the system for multiple days of survey work | Minimum 20 Ah capacity | Must | Achieved | 85Ah battery chosen |
| Battery can deliver enough current to power the DC-DC converter | >30A | Must | Achieved | Battery can supply 30A for almost 1 hour |
| Battery does not discharge to unsafe levels during use | Power supply shuts off when battery voltage is too low | Must | Achieved | EN/UVLO pin on LT3748 switches off at $V_{in}$ = 10V |
| Battery can be recharged easily | Safely charges overnight | Must | Achieved | Simple 12V battery charger required |
| Battery internals are non-hazardous and can be disposed of safely | Prefer non-toxic batteries | Wish | Not Achieved | Flooded lead-acid battery used for commercial reasons |
| Battery can operate in suitable temperature range | -10°C to 40°C | Must | Achieved | Rated for -15°C → 45°C |
| Battery does not need regularly replacing | Expected life >1 year | Must | Achieved | 2 year warranty |
| Battery connections are secure and failsafe | Battery cannot be connected in reverse polarity. | Must | Achieved | Heavy duty Anderson plug specified |
| Battery should be low-cost | Suitable technology in developing countries preferred | Wish | Achieved | Budget option flooded lead-acid technology specified |
| Battery can be carried comfortably by a single person | Less than 15kg | Wish | Not achieved | 21 kg battery |
| **DC-DC Converter** | | | | |
| Can reliably convert battery voltage to required output voltage | Minimum 500V output | Must | Achieved | 530V output at typical load |
| Converter can drive enough power to reach required voltage | 100W output power | Must | Partially achieved | Suitable 93W output power |
| Converter includes external on/off switch | Digitally controlled power on/off | Wish | Achieved | Switch or MOSFET on EN/UVLO pin controls on/off |
| Efficiency is not so low battery power is rapidly depleted | >70% conversion efficiency in normal use | Wish | Partially achieved | At typical loads efficiency is 80%+, drops at high loads |
| Output polarity can be switched to avoid ground polarisation | H-bridge or digitally controlled switching | Wish | Not achieved | Project time constraints meant this could not be completed |
| Safety mechanisms are included to mitigate operator risk | Short circuit protection | Must | Not achieved | Safety regulations should be considered before moving forward with prototyping |
| Electromagnetic Interference (EMI) emissions are minimised | EMI shielding on key parts of the system | Wish | Not achieved | EMI emission standards checked at prototyping stage |
| Output voltage ripple is minimised | <1V ripple | Wish | Achieved | 0.9V output ripple |

The only '**Must**' requirement 'not achieved' was safety mechanisms in the converter. This was not met due to time constraints on the project as there was no time for prototyping.

## 3.2. Future Work and Improvements

Due to the changing scope of the project midway through, a number of work packages were left incomplete at the end. These are summarised below.

### 3.2.1. H-bridge Polarity Switching

To avoid polarising the ground by repeatedly applying voltage in one direction, the polarity must be switched. This can be done with a H-bridge circuit (Figure 47). Closing diagonally opposite pairs of switches creates 2 possible current paths in different directions through the load.

An example of a high-voltage H-bridge is shown in Figure 48 [20]. It uses MOSFET driving ICs [21] to switch up to 600V bidirectionally through the load. Implementing this on the output of the DC-DC converter would allow Arduino control of the output polarity.

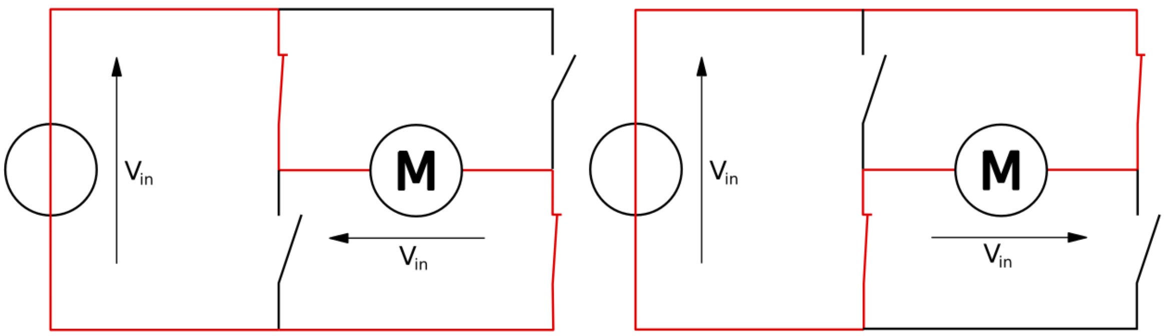



Figure 47 General diagram of a H-bridge showing the 2 possible states achieved by closing switches in pairs. [19]

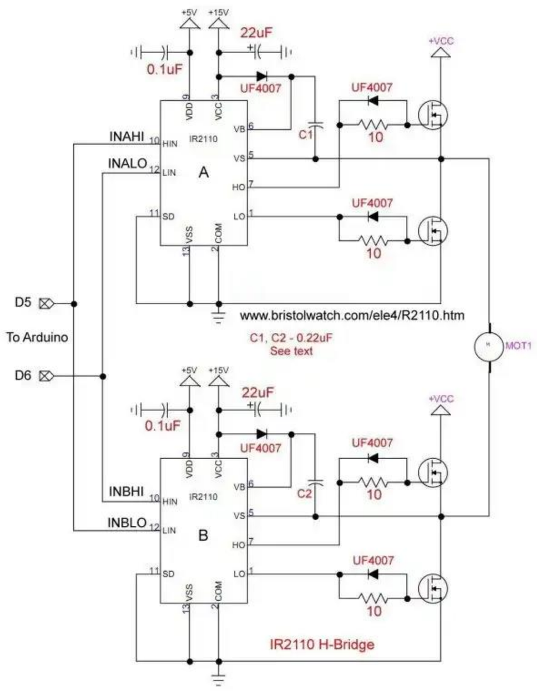



Figure 48 Example of a high voltage H-bridge using MOSFET driver ICs

### 3.2.2. Dynamic Voltage Control

Section 2.1.4.2 demonstrated the ability to vary output voltage using the feedback resistor value. One method of doing this would be to implement digital potentiometer [22] (Figure 49). This could be controlled by an Arduino to vary output voltage according to user inputs.


Figure 49 10kΩ digital potentiometer [22]

### 3.2.3. Converter Alternatives

More modern, complex isolated DC-DC converters are capable of outputting more power than a flyback converter [23][24]. Initially these were rejected due to the increased complexity and adequacy of flyback converters, however in order to develop more power for deeper surveys a different converter topology will likely be required.

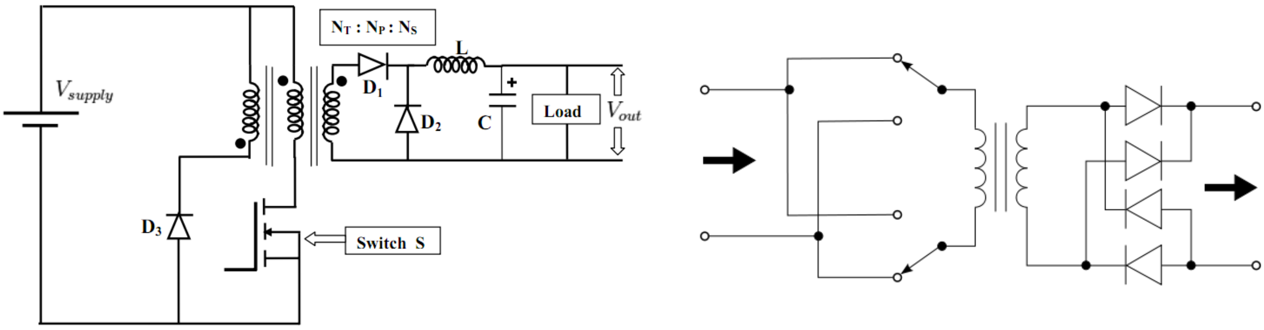



Figure 50 (left) Forward converter capable of up to 200W, (right) push-pull topology capable of up to 1000W

### 3.2.4. Lithium-ion Battery Pack

Rejected due to cost and increased complexity, LFP batteries are far superior battery technology to lead-acid. A custom designed Lithium-ion battery pack would save considerable weight and would need recharging less.

This is a sizeable work package as it could include designing a custom battery management system with charging protection.

Figure 51 Example of a custom LFP battery pack [instructables]

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# Appendices

## Appendix A Regulator Selection Table

| Part Number | | | Vin (mi | Vin (ma | Output | Topology | Mo | Rating /10 | Comments | 1ku Lis | Package |
|---|---|---|---|---|---|---|---|---|---|---|---|
| LT1725 | # | 1 | 16 | 75 | 50 | Flyback | No | | General Purpose Isolated Flyback Controller | $3.64 (LT | 16-Lead SOIC (Narrow 0.15 Inch),16 |
| LT1952 | # | 1 | 15.75 | 25 | 300 | Flyback, Forward | No | | Single Switch Synchronous Forward Controller, LT1952: 460µA; VIN On | $3.96 (LT | 16-Lead SSOP (Narrow 0.15 Inch) |
| LTC3806 | 0 | 1 | 10 | 75 | 60 | Flyback | No | | Synchronous Flyback DC/DC Controller | $4.12 (LT | 12-Lead DFN (4mm x 3mm w/ EP) |
| LT3751 | # | 1 | 5 | 24 | 60 | Capacitor Charger, Flyback | No | 8 | Capacitor Charger Controller with Regulation | $4.37 (LT | 20-Lead TSSOP w/ EP,20-Lead QFN ( |
| LT3750 | # | 1 | 3 | 24 | 60 | Capacitor Charger, Flyback | No | 8 | Capacitor Charger Controller | $4.19 (LT | 10-Lead MSOP |
| LT3750A | # | 1 | 3 | 24 | 60 | Capacitor Charger, Flyback | No | 8 | Capacitor Charger Controller | $4.12 (LT | 10-Lead MSOP |
| LT8357 | 0 | 1 | 3 | 60 | 10 | Boost, Flyback, Sepic | No | 8 | Multitopology Controller | $1.87 (LT | 12-Lead MSOP w/ EP |
| LT3757A | # | 1 | 2.9 | 40 | 50 | Boost, Flyback, Inverting, Sepic | No | 8 | Multi-Topology Controller - Improved Load Transient Over LT3757 | $2.47 (LT | 10-Lead MSOP w/ EP,10-Lead DFN ( |
| LT3758A | # | 1 | 5.5 | 100 | 50 | Boost, Flyback, Inverting, Sepic | No | 8 | Multi-Topology Controller - Improved Load Transient Over LT3758 | $2.84 (LT | 10-Lead MSOP w/ EP,10-Lead DFN ( |
| LT3748 | # | 1 | 5 | 100 | 80 | Flyback, No-Opto Flyback | No | 8 | 100V Isolated Flyback Controller | $3.75 (LT | 16(12)-Lead MSOP |
| LT8306 | # | 1 | 4.5 | 60 | 60 | Flyback, No-Opto Flyback | No | 7 | 60V Low IQ No-Opto Isolated Flyback Controller | $0.79 (LT | 6-Lead TSOT-23 |
| LT1950 | # | 1 | 3 | 25 | 300 | Flyback, Forward | No | 7 | Boost, flyback and forward controller. transformer and MOSFET protec | $3.49 (LT | 16-Lead SSOP (Narrow 0.15 Inch) |
| LT3798 | # | 1 | 10 | 38 | 80 | Flyback, No-Opto Flyback | No | 6 | Isolated No Opto-Coupler Flyback Controller with Active PFC | $2.72 (LT | 16-Lead MSOP w/ EP |
| MAX5070 | # | 1 | 11 | 1k | | Flyback, Single-Ended Forward | No | 6 | Single-Ended Current-Mode PWM Controller | $1.31 (M | Small-Outline IC, Narrow (0.15in),M |
| LTC3803 | # | 1 | 9.2 | 75 | 15 | Flyback | No | 5 | SOT-23 Flyback Controller, LTC3803 8.7V Turn On Voltage, 200kHz | $1.38 (LT | 6-Lead TSOT-23 |
| LTC3803-3 | # | 1 | 9.2 | 75 | 15 | Flyback | No | 5 | SOT-23 Flyback Controller, LTC3803-3 8.7V Turn On Voltage, 300kHz | $1.38 (LT | 6-Lead TSOT-23 |
| LTC3803-5 | # | 1 | 5.7 | 75 | 15 | Flyback | No | 5 | SOT-23 Flyback Controller, LTC3803-5 4.8V Turn On Voltage, 200kHz | $1.38 (LT | 6-Lead TSOT-23 |
| LTC3805 | # | 1 | 8.8 | 75 | 50 | Boost, Flyback, Sepic | No | 5 | Adjustable Frequency Current Mode Flyback, LTC3805 8.4V Turn On Vo | $1.71 (LT | 10-Lead MSOP w/ EP,10-Lead DFN ( |
| LTC3873 | # | 1 | 8.8 | 75 | 50 | Boost, Flyback, Sepic | No | 5 | No RSENSE Boost, LTC3873 9V to 75V During Start-Up | $1.53 (LT | 8-Lead SOT-23,8-Lead DFN (3mm x 2 |
| LTC3805-5 | # | 1 | 4.5 | 75 | 50 | Boost, Flyback | No | 5 | Adjustable Frequency Current Mode Flyback, LTC3805-5 4.5V Turn On \ | $1.71 (LT | 10-Lead MSOP w/ EP,10-Lead DFN ( |
| LTC3873-5 | # | 1 | 4 | 75 | 50 | Boost, Flyback, Sepic | No | 5 | No RSENSE Boost, LTC3873-5 5V to 75V During Start-Up | $1.53 (LT | 8-Lead SOT-23,8-Lead DFN (3mm x 2 |
| MAX15001 | | 1 | 9.5 | 24 | | Flyback, Single-Ended Forward | No | 5 | Current-Mode PWM Controller with Programmable Switching Frequen | $1.64 (M | 10-MINI_SO-N/A |
| MAX15005D | # | 1 | 4.5 | 40 | | Boost, Flyback, Sepic | No | 5 | 4.5V to 40V Input Automotive Flyback/Boost/SEPIC Power-Supply Cont | $2.39 (M | 16-TSSOP_4.4-4.4_MM |
| LT3837 | 0 | 1 | 4.5 | 20 | 80 | Flyback, No-Opto Flyback | No | 4 | Isolated No-Opto Synchronous Flyback Controller | $3.72 (LT | 16-Lead TSSOP w/ EP |
| MAX5095 | | 1 | 8.4 | 26.5 | | Flyback, Single-Ended Forward | No | 4 | High-Performance Single-Ended, Current-Mode PWM Controller | $1.08 (M | Micro-MAX, Thin Shrink Small-Outli |
| LT1737 | # | 1 | 5 | 22 | 25 | Flyback | No | 3 | High Power Isolated Flyback Controller | $3.64 (LT | 16-Lead SOIC (Narrow 0.15 Inch),16 |
| LTC1871 | # | 1 | 2.5 | 36 | 50 | Boost, Flyback, Inverting, Sepic | No | 2 | No RSENSE Current Mode Boost, Flyback and SEPIC Controller | $3.19 (LT | 10-Lead MSOP |
| LTC1871-1 | # | 1 | 2.5 | 36 | 50 | Boost, Flyback, Inverting, Sepic | No | 2 | No RSENSE Current Mode Boost, LTC1871-1 Lower Burst Mode Threshc | $3.19 (LT | 10-Lead MSOP |
| LTC1871-7 | # | 1 | 2.5 | 36 | 50 | Boost, Flyback, Inverting, Sepic | No | 2 | LTC1871-7 Drives 6V-gate N-Channel MOSFETs | $3.19 (LT | 10-Lead MSOP |
| LTC1871X | 2 | 1 | 2.5 | 36 | 50 | Boost, Flyback, Inverting, Sepic | No | 2 | Wide Input Range, Boost, Flyback and SEPIC Controller, 175C Max Junct | $147.20 | 10-Lead MSOP |
| LT1952-1 | # | 1 | 8.3 | 25 | 300 | Flyback, Forward | No | 2 | Single Switch Synchronous Forward Controller, LT1952-1: 400µA; VIN C | $3.96 (LT | 16-Lead SSOP (Narrow 0.15 Inch) |
| LT8471 | # | 2 | 2.6 | 50 | 20 | Boost, Buck, Flyback, Inverting, Sepic, Zeta | Yes | 1 | Dual, Multitopology (Buck/Boost/SEPIC/Inverter/Flyback) | $4.48 (LT | 20-Lead TSSOP w/ EP,28-Lead QFN ( |
| LT8315 | # | 1 | 18 | 560 | 15 | Flyback, No-Opto Flyback | Yes | | 560VIN Micropower No-Opto Isolated Flyback Converter with 630V/30 | $4.90 (LT | 20-Lead TSSOP w/ EP |
| LT3825 | 0 | 1 | 14 | 75 | 80 | Flyback, No-Opto Flyback | No | | Isolated No-Opto Synchronous Flyback Controller with Wide Input Supp | $3.72 (LT | 16-Lead TSSOP w/ EP |
| LT8316 | # | 1 | 16 | 600 | 100 | Buck, Flyback, No-Opto Flyback | No | | 600Vin Micropower No-Opto Isolated Flyback Controller | $2.57 (LT | 20-Lead TSSOP w/ EP,TSSOP 4.4 MN |
| LT8309 | # | 1 | 4.5 | 40 | 80 | Flyback Synchronous Rectifier | No | | Secondary-Side Synchronous Rectifier Driver | $2.16 (LT | 5-Lead TSOT-23 |

# Appendix B MOSFET Selection Tables

| Mfr Part Number | Mfr. | Datasheet | Availability | Pricing | R | | Product Detail | Id - Contin | Qg - Gate Charge | PINTVcc (W | Rds On - Drai | Rdson * Qg: fig of merit (FOM) | Vds - I |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| IPP076N15N5AKSA1 | Infineon | https://www.mouse | 2,745 In Stock | £4.99 | RoHS C | | https://www.mo | 112 A | 61 nC | 0.024 | 5.9 mOhms | 359.9 | 150 V |
| IPI076N15N5AKSA1 | Infineon | https://www.mouse | 12 In Stock | £4.89 | RoHS C | | https://www.mo | 112 A | 61 nC | 0.024 | 5.9 mOhms | 359.9 | 150 V |
| NTP7D3N15MC | onsemi | https://www.mouse | 1,273 In Stock | £3.66 | RoHS C | | https://www.mo | 101 A | 53 nC | 0.021 | 7.3 mOhms | 386.9 | 150 V |
| FDP083N15A-F102 | onsemi / | https://www.mouse | 5,763 In Stock | £4.20 | RoHS C | | https://www.mo | 105 A | 64.5 nC | 0.026 | 6.85 mOhms | 441.825 | 150 V |
| IPP200N15N3 G | Infineon | https://www.mouse | 1,832 In Stock | £2.73 | RoHS C | | https://www.mo | 50 A | 23 nC | 0.009 | 20 mOhms | 460 | 150 V |
| IPP200N15N3GXKSA1 | Infineon | https://www.mouse | 10,965 In Stock | £2.73 | RoHS C | | https://www.mo | 50 A | 31 nC | 0.012 | 16 mOhms | 496 | 150 V |
| IPA105N15N3GXKSA1 | Infineon | https://www.mouse | 21 In Stock | £4.61 | RoHS C | | https://www.mo | 37 A | 55 nC | 0.022 | 9.1 mOhms | 500.5 | 150 V |
| IPA105N15N3 G | Infineon | https://www.mouse | 56 In Stock | £4.52 | RoHS C | | https://www.mo | 37 A | 55 nC | 0.022 | 9.1 mOhms | 500.5 | 150 V |
| IPP111N15N3GXKSA1 | Infineon | https://www.mouse | 502 In Stock | £4.95 | RoHS C | | https://www.mo | 83 A | 55 nC | 0.022 | 9.4 mOhms | 517 | 150 V |
| IPP111N15N3 G | Infineon | https://www.mouse | 395 In Stock | £4.74 | RoHS C | | https://www.mo | 83 A | 55 nC | 0.022 | 9.4 mOhms | 517 | 150 V |
| FDPF190N15A | onsemi / | https://www.mouse | 2,706 In Stock | £2.58 | RoHS C | | https://www.mo | 27.4 A | 39 nC | 0.016 | 14.7 mOhms | 573.3 | 150 V |
| IPP120N20NFDAKSA1 | Infineon | https://www.mouse | 2,712 In Stock | £6.04 | RoHS C | | https://www.mo | 84 A | 65 nC | 0.026 | 10.6 mOhms | 689 | 200 V |
| IPP120N20NFD | Infineon | https://www.mouse | 1,615 In Stock | £6.04 | RoHS C | | https://www.mo | 84 A | 65 nC | 0.026 | 10.6 mOhms | 689 | 200 V |
| IRFB4321PBF | Infineon | | 9,170 In Stock | £2.88 | RoHS C | | https://www.mo | 83 A | 71 nC | 0.028 | 12 mOhms | 852 | 150 V |
| IXTP86N20X4 | IXYS | https://www.mouse | 689 In Stock | £10.24 | RoHS C | | https://www.mo | 86 A | 70 nC | 0.028 | 13 mOhms | 910 | 200 V |
| IXFQ60N25X3 | IXYS | https://www.mouse | 19 In Stock | £8.00 | RoHS C | | https://www.mo | 60 A | 50 nC | 0.020 | 19 mOhms | 950 | 250 V |
| IXFP60N25X3M | IXYS | https://www.mouse | 47 In Stock | £7.25 | RoHS C | | https://www.mo | 60 A | 50 nC | 0.020 | 19 mOhms | 950 | 250 V |
| IXFQ72N20X3 | IXYS | https://www.mouse | 13 In Stock | £8.32 | RoHS C | | https://www.mo | 72 A | 55 nC | 0.022 | 20 mOhms | 1100 | 200 V |
| IXFP72N20X3M | IXYS | https://www.mouse | 126 In Stock | £7.52 | RoHS C | | https://www.mo | 72 A | 55 nC | 0.022 | 20 mOhms | 1100 | 200 V |
| IXFP72N20X3 | IXYS | https://www.mouse | 315 In Stock | £7.37 | RoHS C | | https://www.mo | 72 A | 55 nC | 0.022 | 20 mOhms | 1100 | 200 V |
| TP65H015G5WS | Transphc | https://www.mouse | 666 In Stock | £27.75 | RoH Ne | | https://www.mo | 95 A | 68 nC | 0.027 | 18 mOhms | 1224 | 650 V |
| IPI111N15N3 G | Infineon | https://www.mouse | Non-Stocked Lea | £4.50 | | | https://www.mo | 83 A | 41 nC | 0.016 | 10.8 mOhms | 442.8 | 150 V |
| IPI111N15N3GAKSA1 | Infineon | https://www.mouse | Non-Stocked Lea | £2.82 | RoHS C | | https://www.mo | 83 A | 55 nC | 0.022 | 10.8 mOhms | 594 | 150 V |

# Appendix C Diode Selection Tables

| Mfr Part Number | Mfr | Datashe | Availability | Pricing | Ro | Lifecyc | Product Detail | Vr - Ro | Mounti | Package/Ca | If - F | Type | Configu | Vf - | Max Surge Current | Ir - Reverse | Recovery Time | merit (Vf * rec |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SF1660CT-BP | Micro ( | https://ww | Non-Stocked Lead-Time 15 Weeks | £0.38 | | New Proc | https://www.mo | 600 V | Through Hole | | 16 A | Ultra Fas | Single | 1.4 | 100 A | 10 uA | 35 | 49 |
| MUR460_E | Diodes | https://ww | 415 In Stock | £0.42 | | New At M | https://www.mo | 600 V | Through H | DO-201AD-2 | 4 A | Ultra Fas | Single | 1.25 | 220 A | 10 uA | 50 | 62.5 |
| MUR460-T | Diodes | https://ww | 2,737 In Stock | £0.41 | | New At M | https://www.mo | 600 V | Through H | DO-201AD-2 | 4 A | Ultra Fas | Single | 1.28 | 220 A | 10 uA | 50 | 64 |
| DTH3006PT | Diodes | https://ww | 20 In Stock | £2.24 | | New Proc | https://www.mo | 600 V | Through H | TO247-2 | 30 A | Ultra Fas | Single | 2.4 | 350 A | 100 uA | 27 | 64.8 |
| 1N6622U/TR | Microc | https://ww | Non-Stocked | £7.33 | | New Proc | https://www.mo | 660 V | Through H | A | 2 A | Ultra Fas | Single | 1.6 | 3.5 A | 500 nA | 45 | 72 |
| STPR560D | Diodes | https://ww | 200 In Stock | £0.46 | RoHS | New At M | https://www.mo | 600 V | Through H | TO-220AC-2 | 5 A | Ultra Fas | Common C | 1.5 | 55 A | 10 uA | 50 | 75 |
| STPR860D | Diodes | https://ww | 170 In Stock | £0.56 | RoHS | New At M | https://www.mo | 600 V | Through H | TO-220AC-2 | 8 A | Ultra Fas | Single | 1.5 | 90 A | 10 uA | 60 | 90 |
| DTH810FP | Diodes | https://ww | 104 In Stock | £0.74 | | New Proc | https://www.mo | 1 kV | Through H | ITO220AC-2 | 8 A | Ultra Fas | Single | 2 | 80 A | 5 uA | 48 | 96 |
| 1N6630U | Microc | https://ww | Non-Stocked | £18.10 | | New Proc | https://www.mo | 990 V | Through H | E | 3 A | Ultra Fas | Single | 1.7 | 75 A | 2 uA | 60 | 102 |
| 1N6630U/TR | Microc | https://ww | Non-Stocked | £18.26 | | New Proc | https://www.mo | 990 V | Through H | E | 3 A | Ultra Fas | Single | 1.7 | 75 A | 2 uA | 60 | 102 |
| MUR30120BH-BP | Micro ( | https://ww | Non-Stocked Lead-Time 18 Weeks | £1.28 | | New Proc | https://www.mo | 1.2 kV | Through Hole | | 30 A | Ultra Fas | Single | 1.9 | 300 A | 5 uA | 55 | 104.5 |
| DTH3006D | Diodes | https://ww | 1,000 Factory Stock Available | £1.04 | RoHS | New Proc | https://www.mo | 600 V | Through H | TO220AC-2 | 30 A | Ultra Fas | Single | 2.4 | 350 A | 100 uA | 45 | 108 |
| DTH3006FP | Diodes | https://ww | 48 In Stock | £1.23 | RoHS | New Proc | https://www.mo | 600 V | Through H | ITO220AC-2 | 30 A | Ultra Fas | Single | 2.4 | 350 A | 100 uA | 45 | 108 |
| 1N6623U/TR | Microc | https://ww | Non-Stocked | £11.36 | | New Proc | https://www.mo | 880 V | Through H | A | 1.5 A | Ultra Fas | Single | 1.8 | 4.2 A | 500 nA | 60 | 108 |
| UF4007GP-AP | Micro ( | https://ww | 7,666 In Stock | £0.22 | RoHS | New Proc | https://www.mo | 1 kV | Through H | DO-41 | 1 A | Ultra Fas | Single | 1.7 | 30 A | 5 uA | 75 | 127.5 |
| 1N6631E3 | Microc | https://ww | Non-Stocked | £9.82 | | New Proc | https://www.mo | 1.1 kV | Through H | E | 4 A | Ultra Fas | Single | 1.6 | 60 A | 4 uA | 80 | 128 |
| 1N6631E3/TR | Microc | https://ww | Non-Stocked | £9.99 | | New Proc | https://www.mo | 1.1 kV | Through H | E | 4 A | Ultra Fas | Single | 1.6 | 60 A | 4 uA | 80 | 128 |
| 1N6631U | Microc | https://ww | Non-Stocked | £15.54 | | New Proc | https://www.mo | 1.1 kV | Through H | E | 4 A | Ultra Fas | Single | 1.6 | 60 A | 4 uA | 80 | 128 |
| 1N6631U/TR | Microc | https://ww | Non-Stocked | £15.70 | | New Proc | https://www.mo | 1.1 kV | Through H | E | 4 A | Ultra Fas | Single | 1.6 | 60 A | 4 uA | 80 | 128 |
| UF4010G_AY_00001 | Panjit | https://ww | 2,901 In Stock | £0.36 | RoHS | Compliar | https://www.mo | 1 kV | Through H | DO-201AD | 4 A | Ultra Fas | Single | 1.85 | 150 A | 10 uA | 75 | 138.75 |
| MUR60120BH-BP | Micro ( | https://ww | Non-Stocked Lead-Time 14 Weeks | £1.83 | | New Proc | https://www.mo | 1.2 kV | Through Hole | | 60 A | Ultra Fas | Single | 2.6 | 500 A | 5 uA | 60 | 156 |

# Appendix D Example VES Data

| | | | | | | | | | | | | | | | | % of input measured | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ax | Bx | AB | Mx | Nx | MN | Fz | IAB(mA) | VAB(V) | VMN(V) | RMN(ohm) | Var% | Rho(Ohm.m) | | Output Power (W) | RAB(ohm) | RMN/RAB (%) | MN/AB |
| 1 | -1 | **2** | 0.2 | -0.2 | **0.4** | 0.37 | 239.00 | 376.3 | **4.295** | 17.9 | 0.09 | 137.2 | | 89.9 | 1574 | 1.1387 | 0.20 |
| 1.4 | -1.4 | **2.8** | 0.2 | -0.2 | **0.4** | 0.53 | 239.00 | 326.4 | **2.474** | 10.32 | 0.13 | 155.7 | | 78.0 | 1366 | 0.7559 | 0.14 |
| 1.9 | -1.9 | **3.8** | 0.2 | -0.2 | **0.4** | 0.72 | 239.00 | 285.3 | **1.500** | 6.262 | 0.19 | 175.5 | | 68.2 | 1194 | 0.5247 | 0.11 |
| 2.7 | -2.7 | **5.4** | 0.2 | -0.2 | **0.4** | 1.03 | 239.00 | 332.9 | **0.822** | 3.432 | 0.20 | 195.4 | | 79.6 | 1393 | 0.2464 | 0.07 |
| 3.7 | -3.7 | **7.4** | 0.2 | -0.2 | **0.4** | 1.41 | 239.00 | 503.3 | **0.492** | 2.055 | 0.16 | 220.3 | | 120.3 | 2106 | 0.0976 | 0.05 |
| 5.2 | -5.2 | **10.4** | 0.2 | -0.2 | **0.4** | 1.99 | 239.00 | 499.9 | **0.225** | 0.940 | 0.19 | 199.4 | | 119.5 | 2092 | 0.0450 | 0.04 |
| 7.2 | -7.2 | **14.4** | 0.2 | -0.2 | **0.4** | 2.76 | 239.00 | 390.7 | **0.120** | 0.499 | 0.15 | 203.1 | | 93.4 | 1635 | 0.0305 | 0.03 |
| 5.2 | -5.2 | **10.4** | 1 | -1 | **2** | 1.93 | 239.00 | 501.1 | **1.070** | 4.467 | 0.19 | 182.7 | | 119.8 | 2097 | 0.2130 | 0.19 |
| 7.2 | -7.2 | **14.4** | 1 | -1 | **2** | 2.71 | 239.00 | 451.8 | **0.575** | 2.399 | 0.16 | 191.5 | | 108.0 | 1890 | 0.1269 | 0.14 |
| 10 | -10 | **20** | 1 | -1 | **2** | 3.80 | 239.00 | 325.2 | **0.259** | 1.081 | 0.16 | 168.0 | | 77.7 | 1361 | 0.0794 | 0.10 |
| 14 | -14 | **28** | 1 | -1 | **2** | 5.34 | 239.00 | 374.9 | **0.117** | 0.489 | 0.14 | 149.8 | | 89.6 | 1569 | 0.0312 | 0.07 |
| 19 | -19 | **38** | 1 | -1 | **2** | 7.26 | 239.00 | 464.2 | **0.061** | 0.253 | 0.11 | 143.1 | | 110.9 | 1942 | 0.0130 | 0.05 |
| 27 | -27 | **54** | 1 | -1 | **2** | 10.33 | 239.00 | 277.3 | **0.028** | 0.116 | 0.15 | 133.2 | | 66.3 | 1160 | 0.0100 | 0.04 |
| 37 | -37 | **74** | 1 | -1 | **2** | 14.17 | 239.00 | 318.7 | **0.014** | 0.059 | 0.06 | 123.2 | | 76.2 | 1334 | 0.00440 | 0.03 |
| 27 | -27 | **54** | 5.2 | -5.2 | **10.4** | 10.02 | 239.00 | 277.7 | **0.169** | 0.704 | 0.12 | 149.3 | | 66.4 | 1162 | 0.0606 | 0.19 |
| 37 | -37 | **74** | 5.2 | -5.2 | **10.4** | 13.94 | 357.99 | 386.6 | **0.125** | 0.348 | 0.08 | 141.1 | | 138.4 | 1080 | 0.0322 | 0.14 |
| 52 | -52 | **104** | 5.2 | -5.2 | **10.4** | 19.75 | 356.21 | 389.5 | **0.059** | 0.167 | 0.19 | 134.8 | | 138.7 | 1094 | 0.0152 | 0.10 |
| 72 | -72 | **144** | 5.2 | -5.2 | **10.4** | 27.47 | 192.02 | 545.0 | **0.016** | 0.084 | 0.30 | 130.5 | | 104.7 | 2838 | 0.00295 | 0.07 |
| 100 | -100 | **200** | 5.2 | -5.2 | **10.4** | 38.22 | 196.00 | 541.1 | **0.0081** | 0.041 | 2.41 | 124.7 | | 106.0 | 2761 | 0.00150 | 0.05 |
| 140 | -140 | **280** | 5.2 | -5.2 | **10.4** | 53.59 | 201.95 | 543.2 | **0.0040** | 0.020 | 3.42 | 117.4 | | 109.7 | 2690 | 0.000738 | 0.04 |
| 149 | -149 | **298** | 5.2 | -5.2 | **10.4** | 57.04 | 206.13 | 545.3 | **0.0038** | 0.018 | 2.74 | 122.4 | | 112.4 | 2646 | 0.000691 | 0.03 |

# Appendix E Power Supply System Bill of Materials (BOM)

| Bill of Materials | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Sub-System | Part No. | Description | Material | Unit Cost (£) | Qty. | Supplier | Cost (£) | Supplier Part No. |
| **DC-DC Flyback Converter** | 1 | Current sense resistor 0.005 ohms | Electronics | 2.13 | 1 | Farnell | 2.13 | 3791120 |
| | 2 | Through hole resistor, 6.04kohm | Electronics | 0.39 | 3 | Farnell | 1.16 | 1083337 |
| | 3 | Through hole resistor, 249kohm | Electronics | 0.43 | 3 | Farnell | 1.29 | 1083513 |
| | 4 | Power MOSFET, N channel, 150V 50A | Electronics | 2.97 | 1 | Farnell | 2.97 | 1775651 |
| | 5 | Ultrafast diode, 800V, 5A, 40ns | Electronics | 1.10 | 1 | Farnell | 1.10 | 3287343 |
| | 6 | SMPS flyback transformer, 10A, 1.5kV | Electronics | 5.10 | 1 | Farnell | 5.10 | 2458039 |
| | 7 | Power film capacitor, 1µF, 800V | Electronics | 1.74 | 3 | Farnell | 5.22 | 3519081 |
| | 8 | Electrolytic capacitor, 10µF, 25V | Electronics | 0.10 | 4 | Farnell | 0.40 | 8766754 |
| | 9 | Electrolytic capacitor, 47µF, 16V | Electronics | 0.08 | 4 | Farnell | 0.31 | 8767114 |
| | 10 | Electrolytic capacitor, 1µF, 50V | Electronics | 0.06 | 2 | Farnell | 0.11 | 1902924 |
| | 11 | Electrolytic capacitor, 0.1µF, 100V | Electronics | 0.06 | 2 | Farnell | 0.11 | 1902938 |
| | 12 | Banana test connector, 32A, 1kV, red | Electronics | 2.48 | 1 | Farnell | 2.48 | 4064885 |
| | 13 | Banana test connector, 32A, 1kV, black | Electronics | 2.32 | 1 | Farnell | 2.32 | 4064883 |
| | 14 | Banana test conenctor, 32A, 30V, black | Electronics | 0.36 | 2 | Farnell | 0.73 | 1698951 |
| | 15 | Banana test conenctor, 32A, 30V, green | Electronics | 0.26 | 2 | Farnell | 0.51 | 1698955 |
| | 16 | Rocker switch, SPST, illuminated | Electronics | 3.18 | 1 | Farnell | 3.18 | 2910111 |
| **Battery** | 17 | 85Ah Deep Cycle Flooded Battery | Lead-acid battery | 90.00 | 1 | Advanced Battery Supplies | 90.00 | - |
| | 18 | Heavy duty jump leads with Anderson connector | Copper, ABS | 26.39 | 1 | AT Leads & Looms Ltd. | 26.39 | - |
| | 19 | Car Battery Charger 12v | Electronics | 6.90 | 1 | Alibaba | 6.90 | 1600698951835 |
| | | | | | | **TOTAL:** | **152.41** | |